Particle detection and gamma-ray astronomy

SAGI Summer School
ICISE QUY NHON
21st July 2025

Housekeeping's

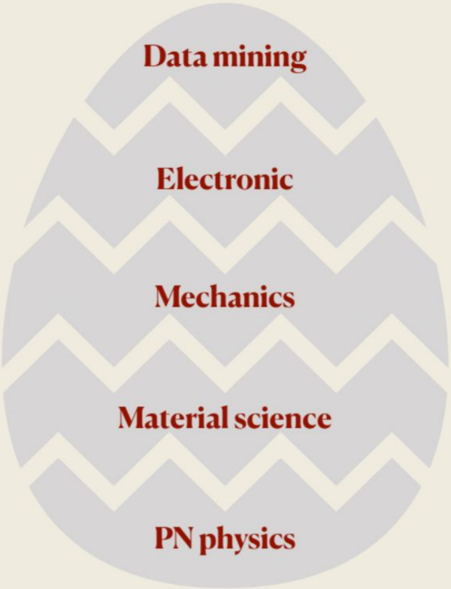
- ★ Close your laptops 🧑‍💻
- ★ Make a name plate 🏷️
- ★ There will be questions during lecture 🙋

OUTLINE

1. Gamma-ray astronomy
   - Gamma-ray astronomy and gamma-ray telescope
   - The Imaging Atmospheric Cherenkov Technique
2. Q&A (1)
3. Beyond Gamma-ray: Cherenkov telescopes as optical telescopes
   - One-pixel telescope
   - Many-pixel telescope
   - Many-telescope telescope
4. Q&A (2)

From Son Cao’s lecture

Particle detector is a *instrument (kind of classical device)* to detect, to track, and to identify the *particles (superposition of quantum states)*



A complicate, interdisciplinary field

“The real voyage of discovery consists, not in seeking new landscapes, but in having new eyes”

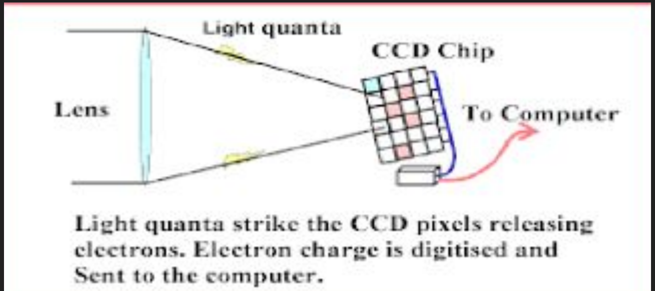
~Marcel Proust~

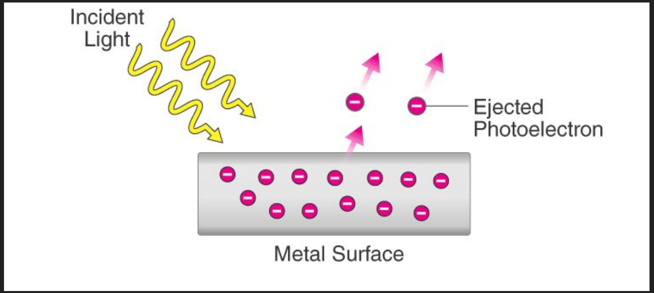
Good luck!

From Son Cao

The optical sky

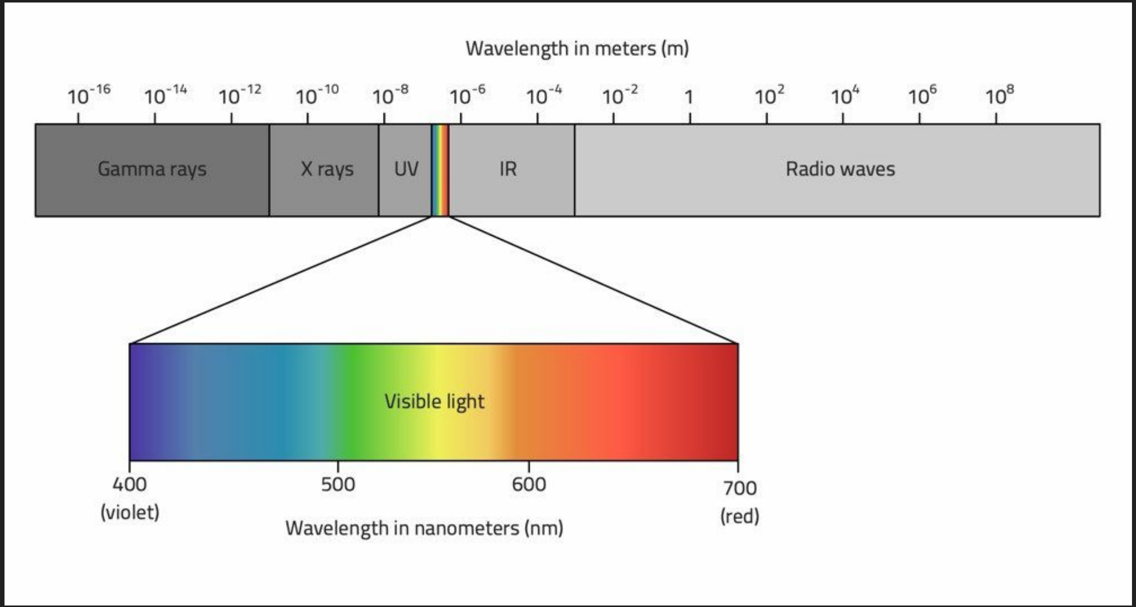
- ★ Question: Is your phone's camera a particle detector? 📸
  - ○ Hint: photoelectric effect

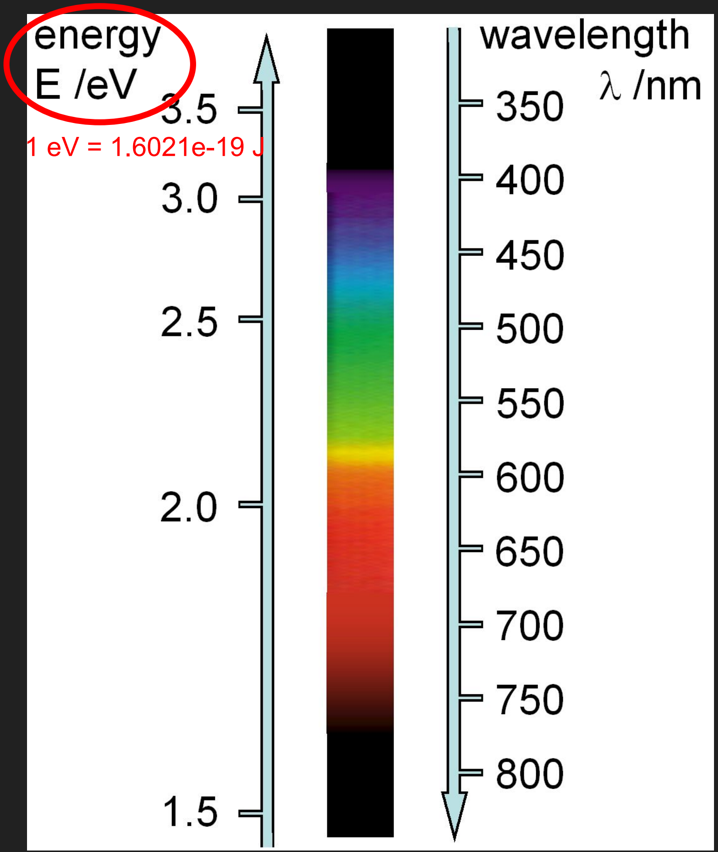




The optical sky

The optical sky





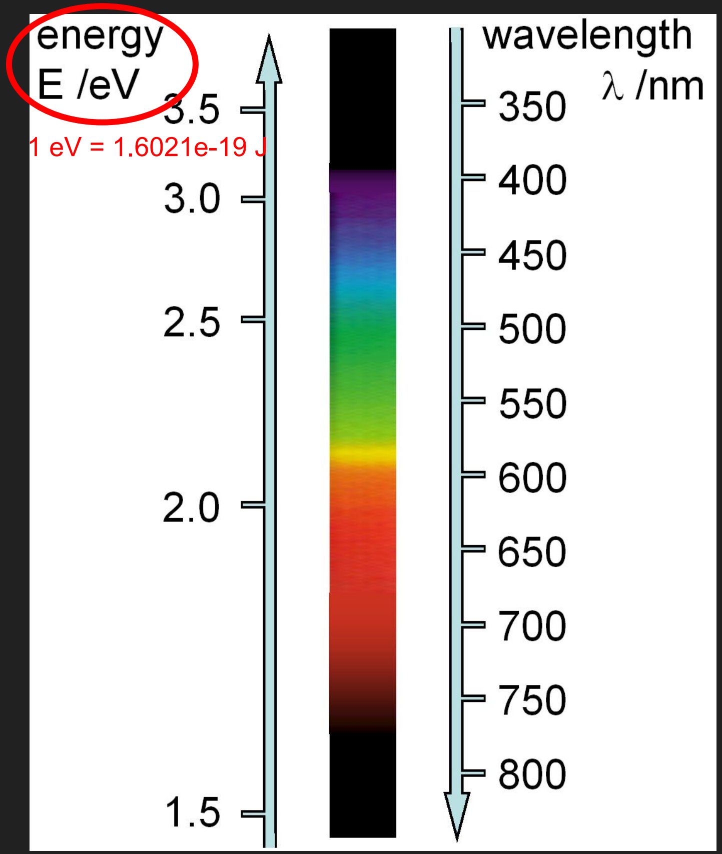
The optical sky

The Relationship between Energy (E), Frequency (υ), Wavelength (λ), and Planck's Constant (h)

$$\boxed{E = h\upsilon} \qquad \boxed{E = \frac{hc}{\lambda}}$$

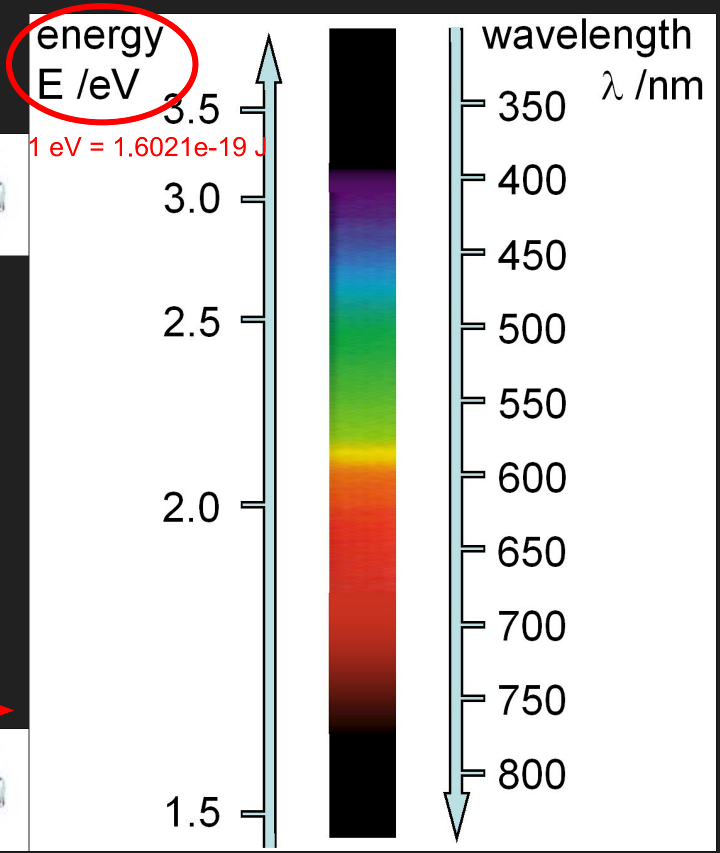
$h = 6.626 \times 10^{-34}$ J·s

$c = 3.0 \times 10^{8}$ m/s

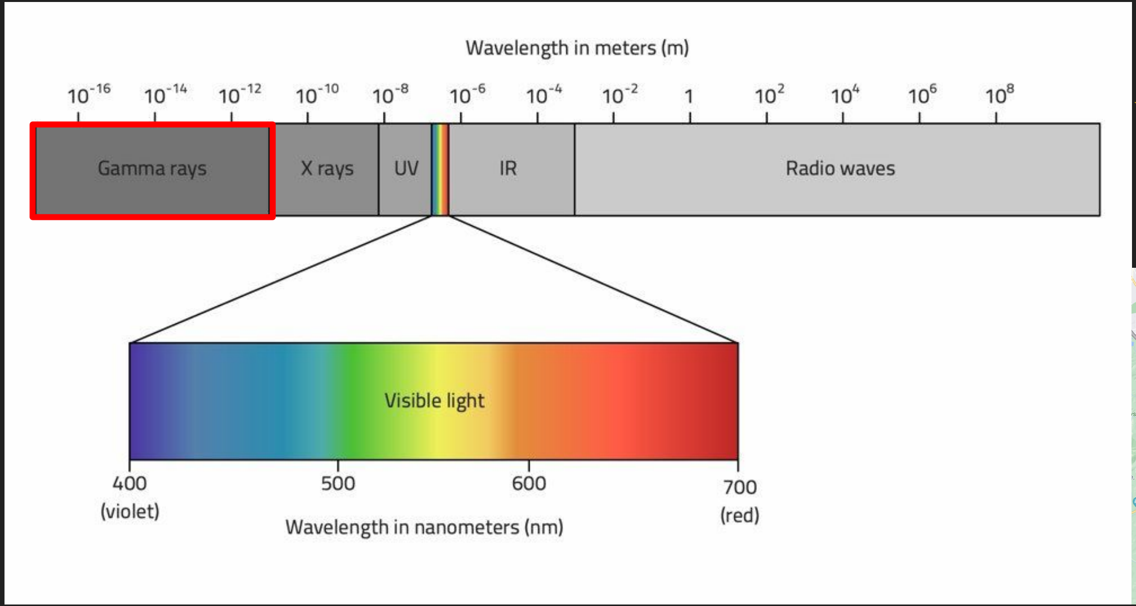


The optical sky

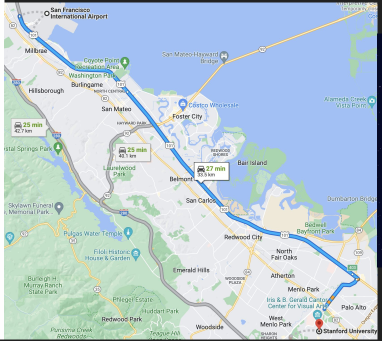
1.5 cm



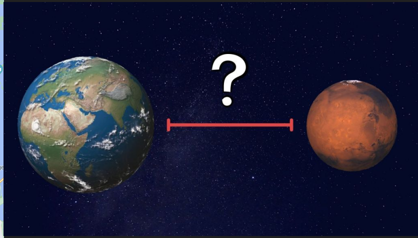
The gamma-ray sky



- ★ Gamma-ray:
  - MeV = a million eV
  - 33 km “battery train”
- ★ Very high energy gamma-ray:
  - TeV = a million MeV = a trillion eV
  - 33 million km

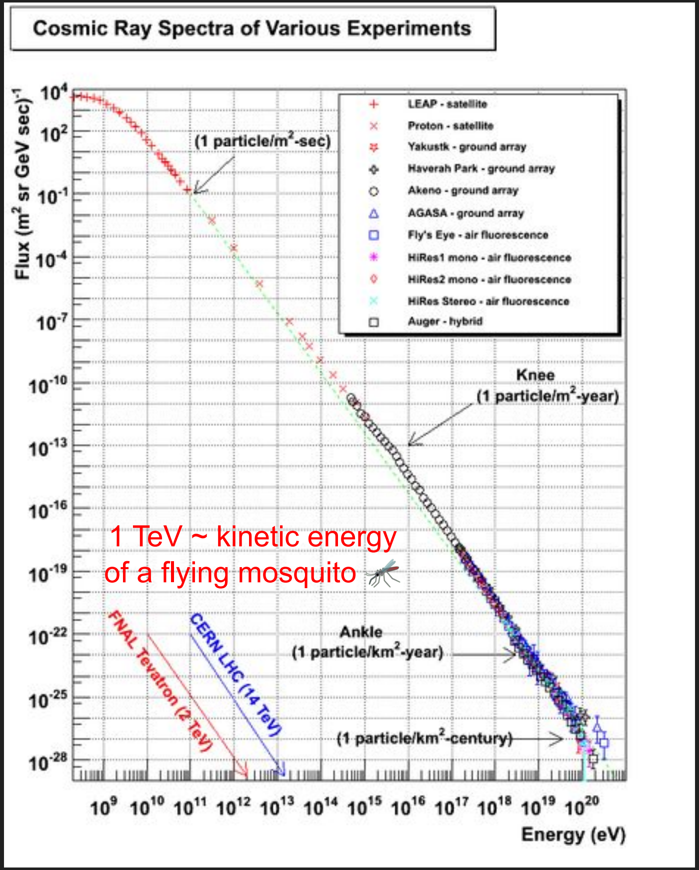


From SFO to Stanford



~ 56 million km

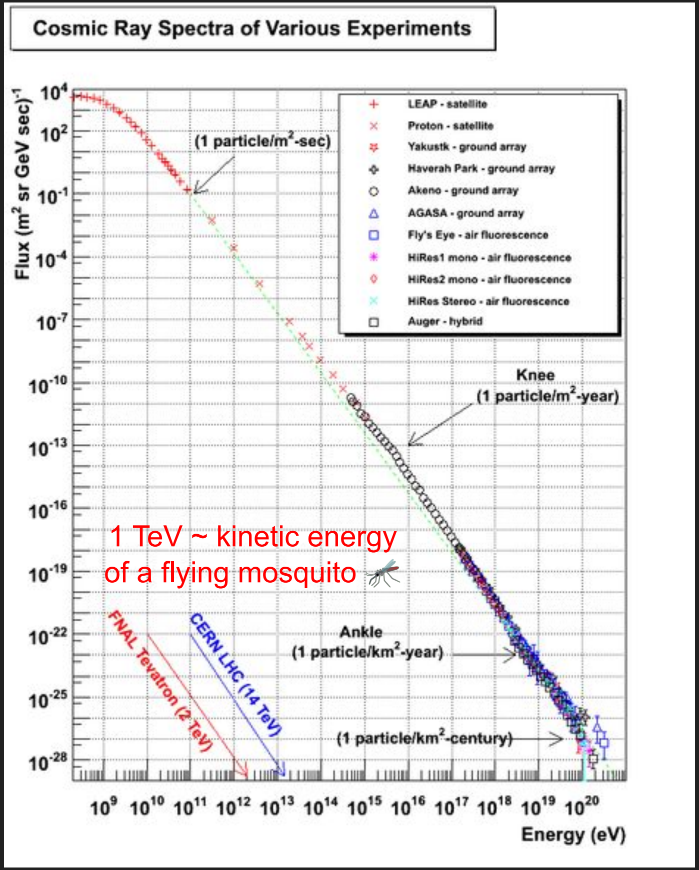
The gamma-ray sky



The gamma-ray sky

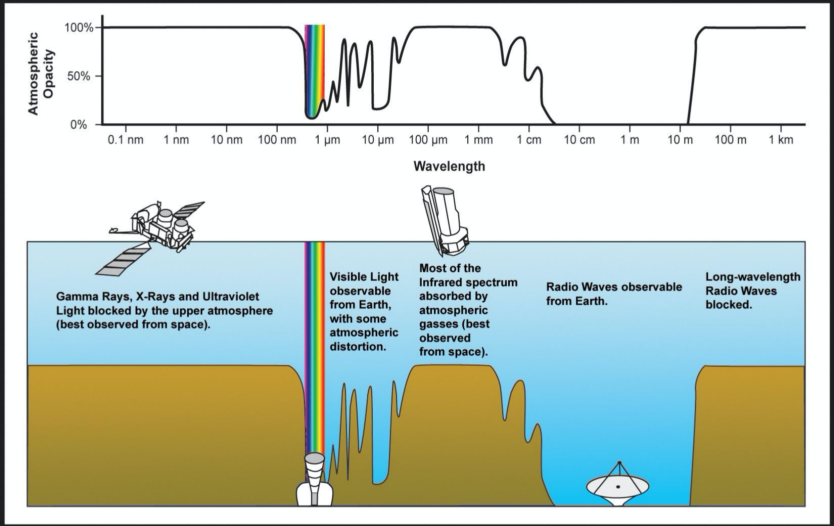
Balloon experiment
1911-1913

Victor Francis Hess,
Nobel Prize in Physics
1936

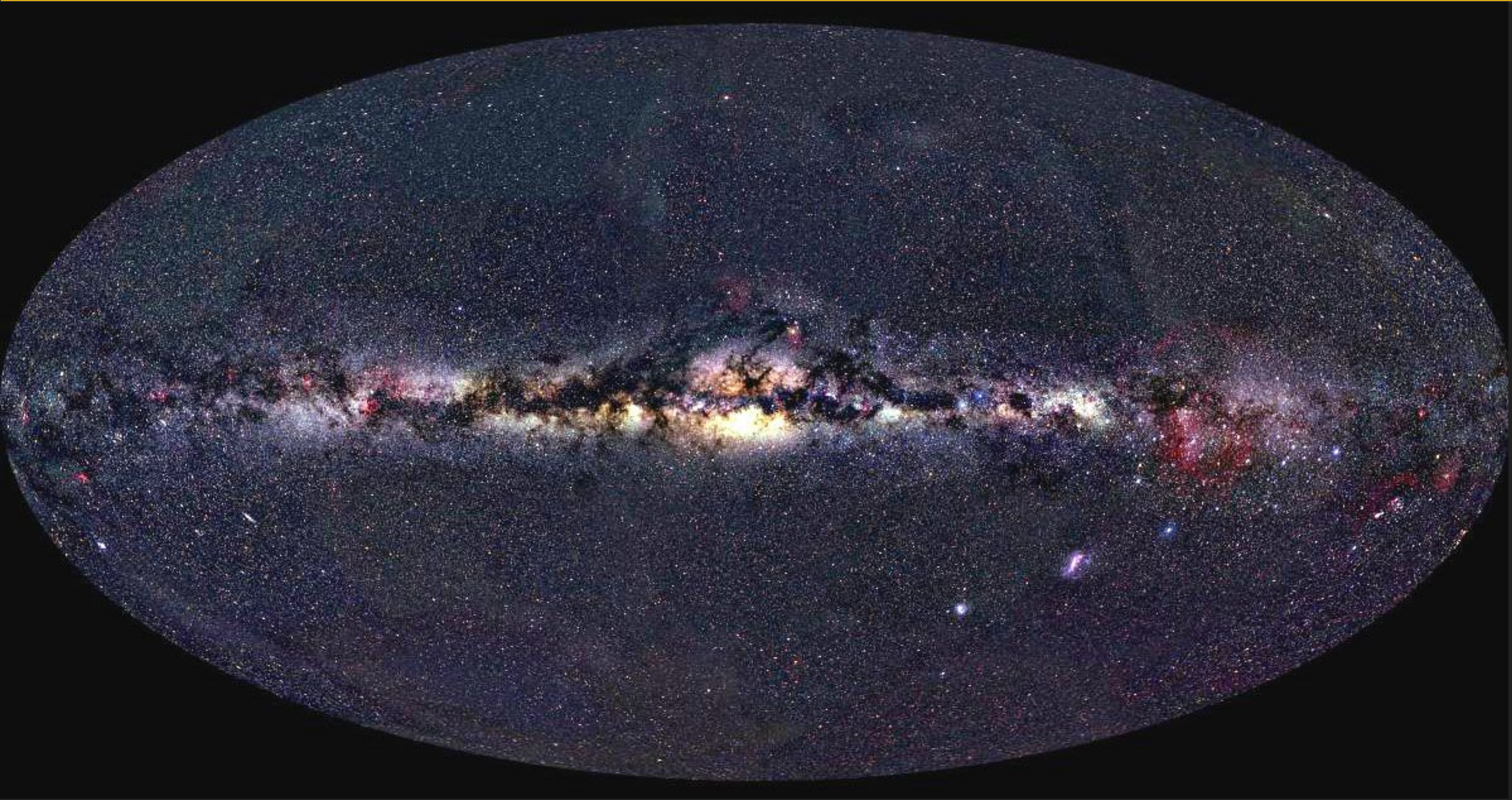


The gamma-ray sky: space or ground?

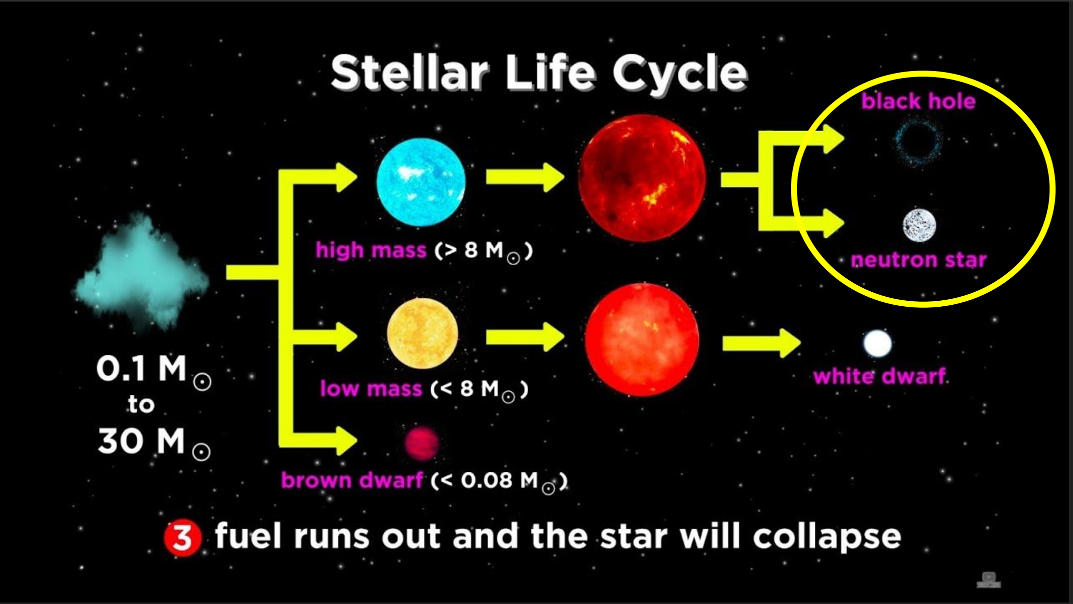
- ★ The Earth's atmosphere is not transparent across the wavelength

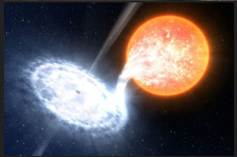


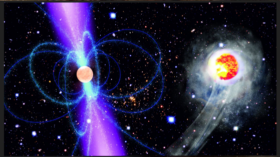
The gamma-ray sky: space or ground?

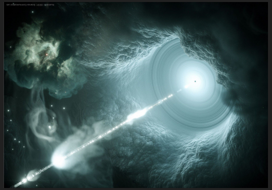


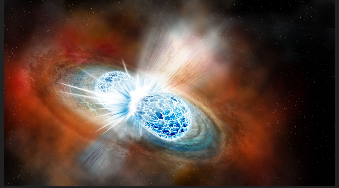
The gamma-ray sky



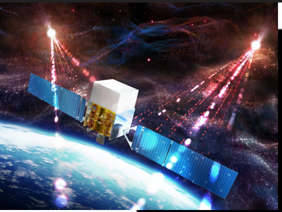


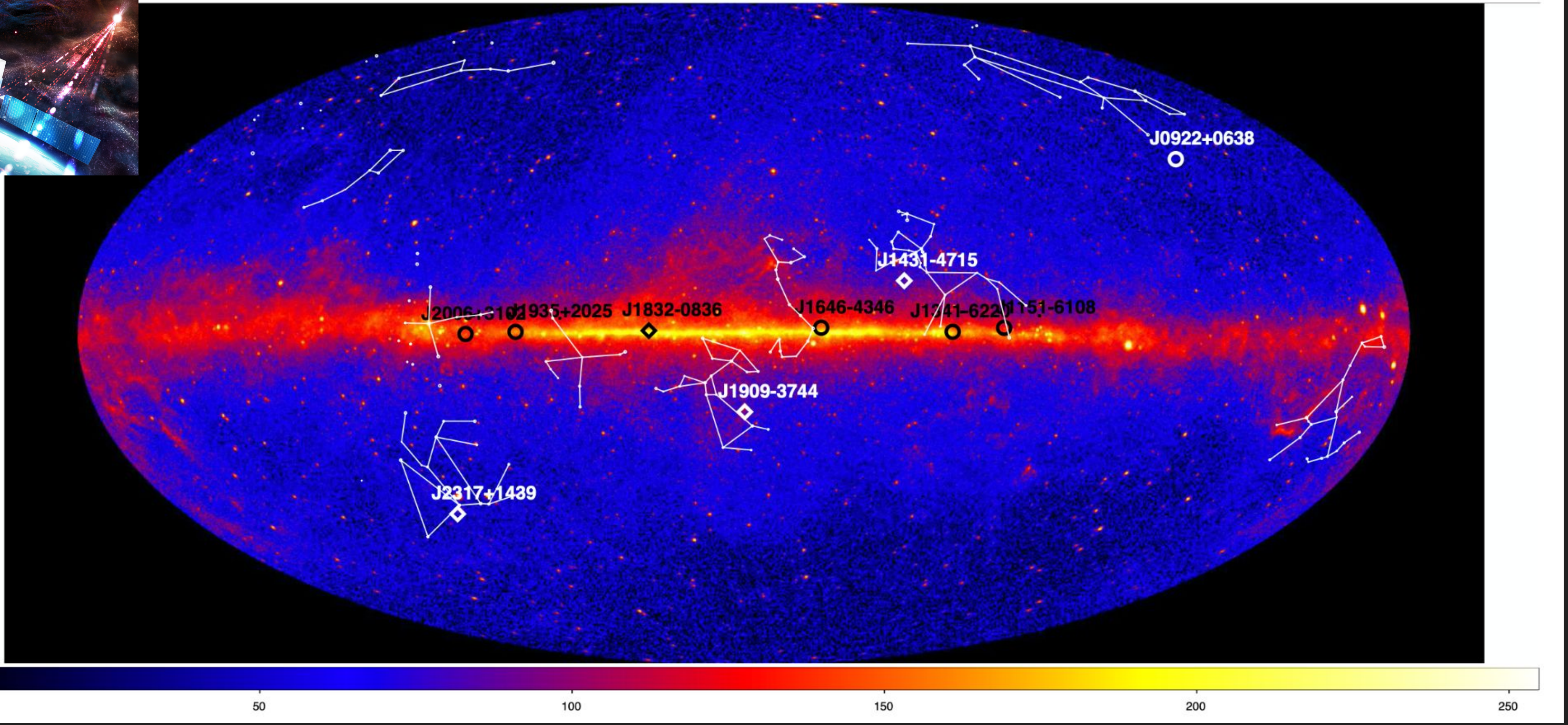






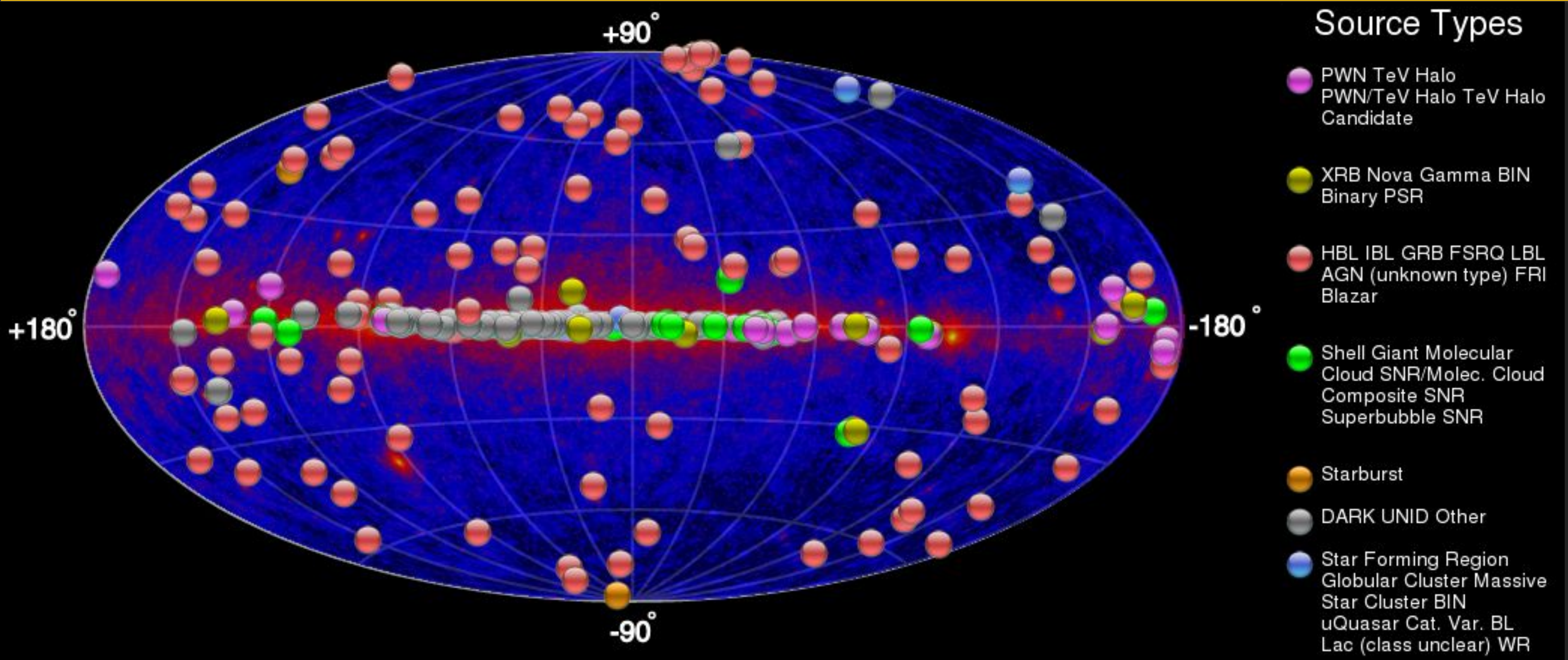
The gamma-ray sky (GeV)





Arxiv: 1706.03592

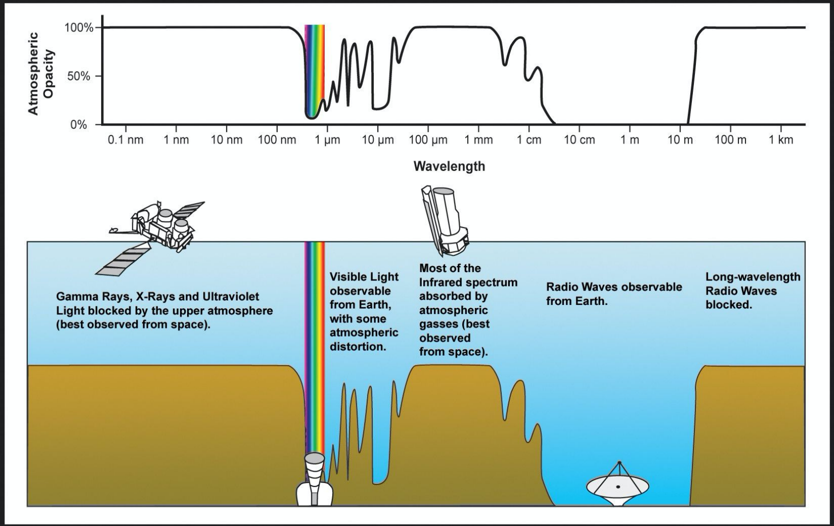
The gamma-ray sky (TeV)



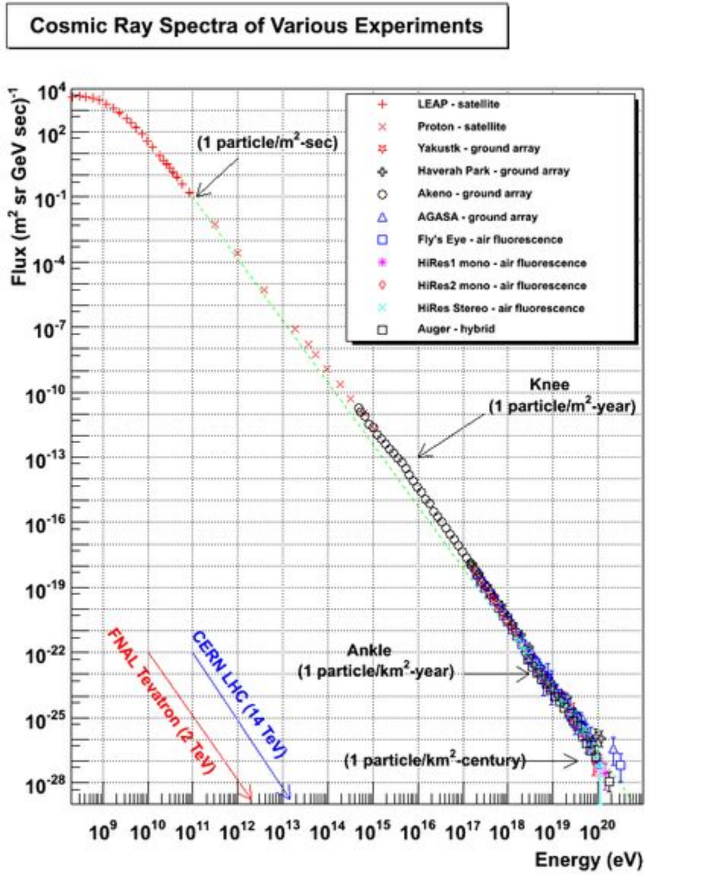
http://tevcat.uchicago.edu

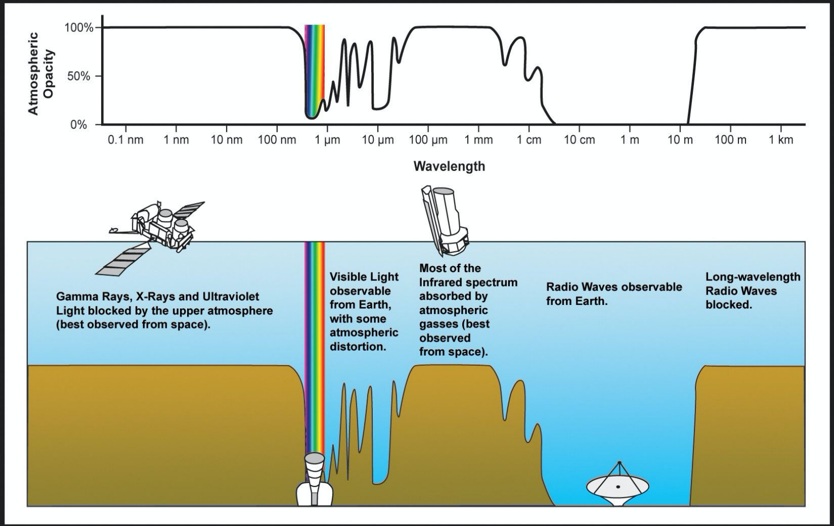
The gamma-ray sky: space or ground?

- ★ Gamma rays allow us to study the extreme universe from astrophysical sources.
- ★ To detect gamma rays:
  - ○ Direct detection: space-borne satellite (*Fermi*)
  - ○ Indirect detection: ground-based Cherenkov Telescopes

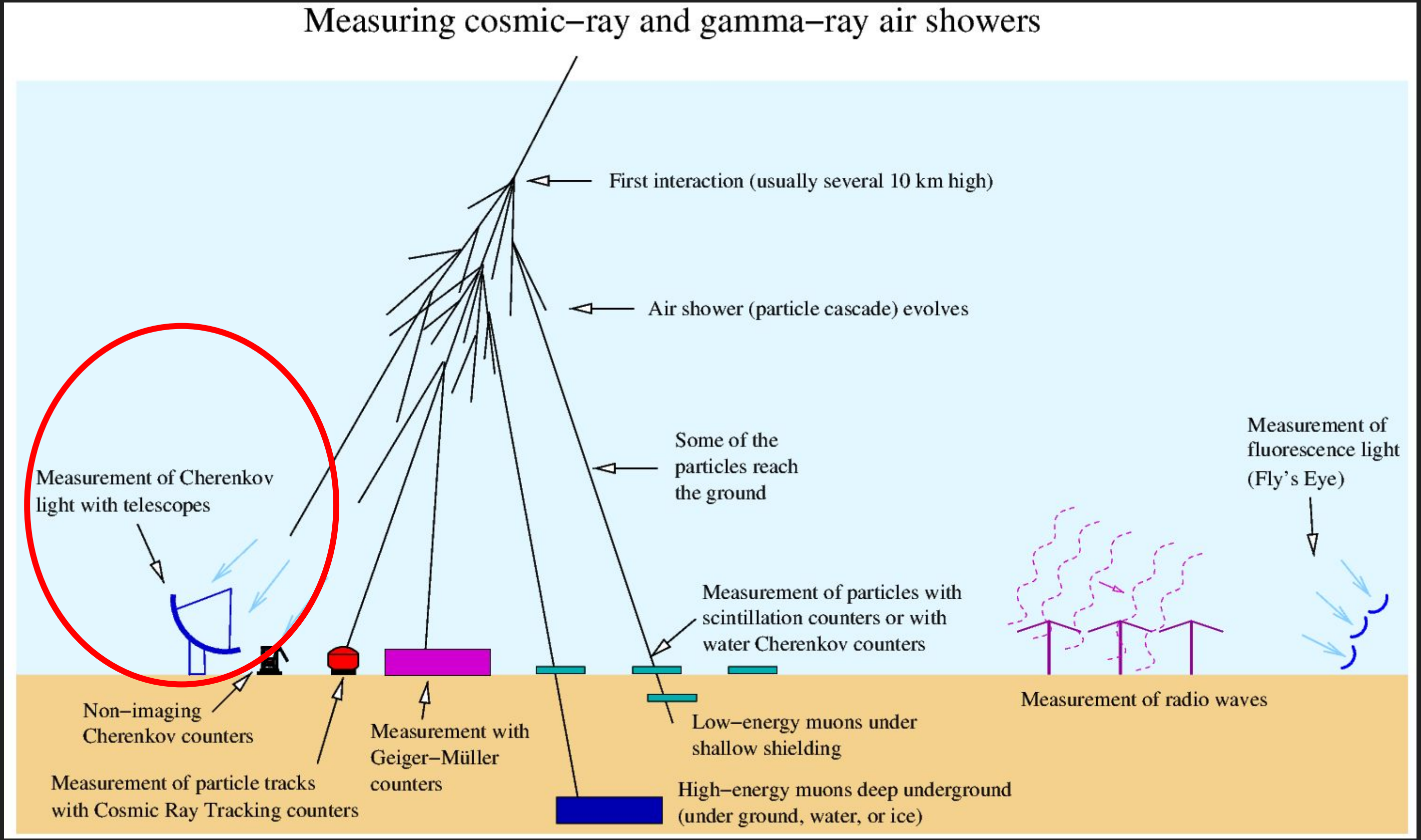


The gamma-ray sky: space or ground?

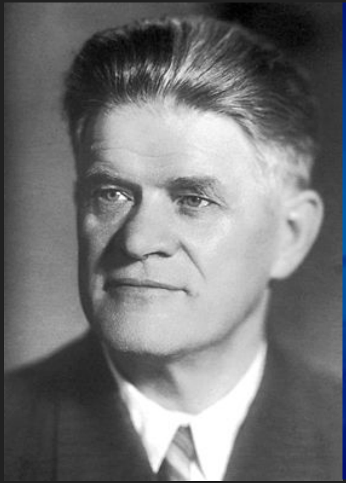




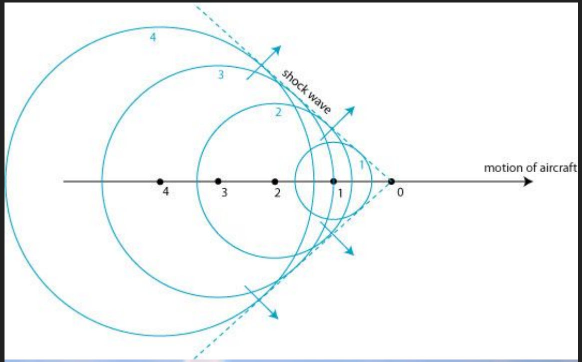
Some methods to detect airshowers



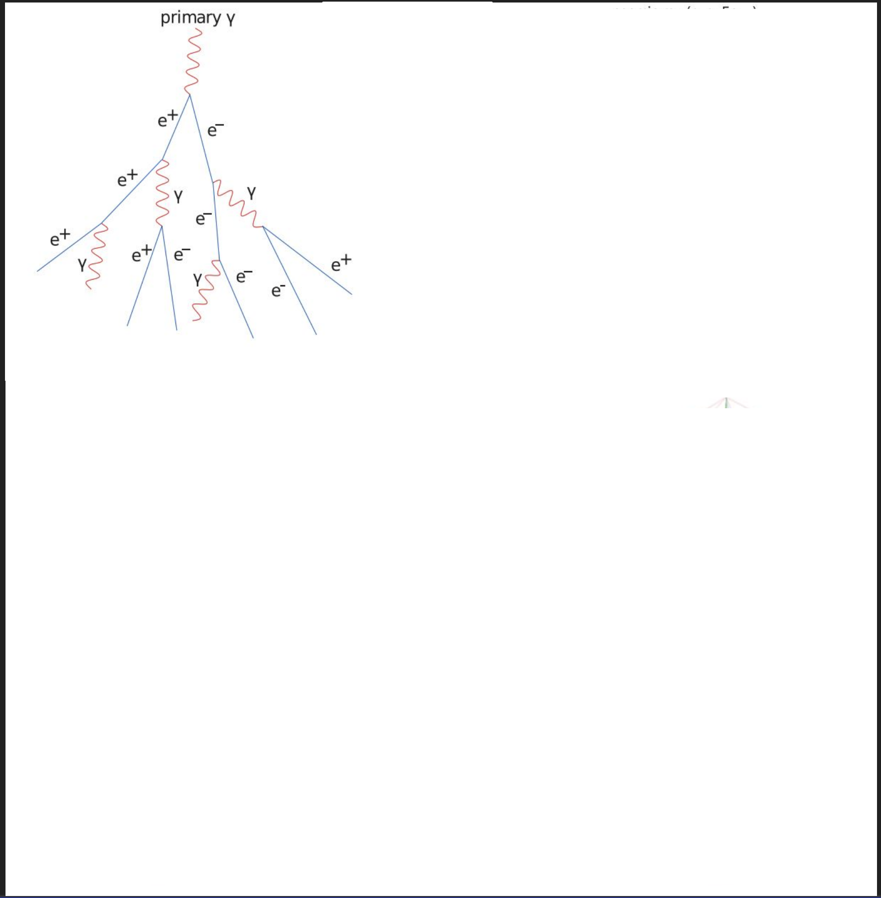
Cherenkov Technique: Cherenkov Radiation

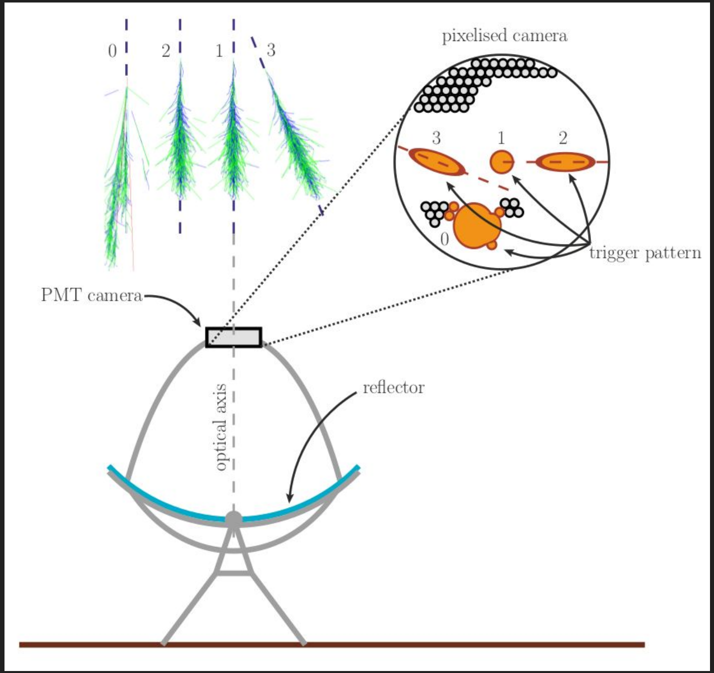



Pavel Cherenkov,
Nobel Prize in
Physics 1958

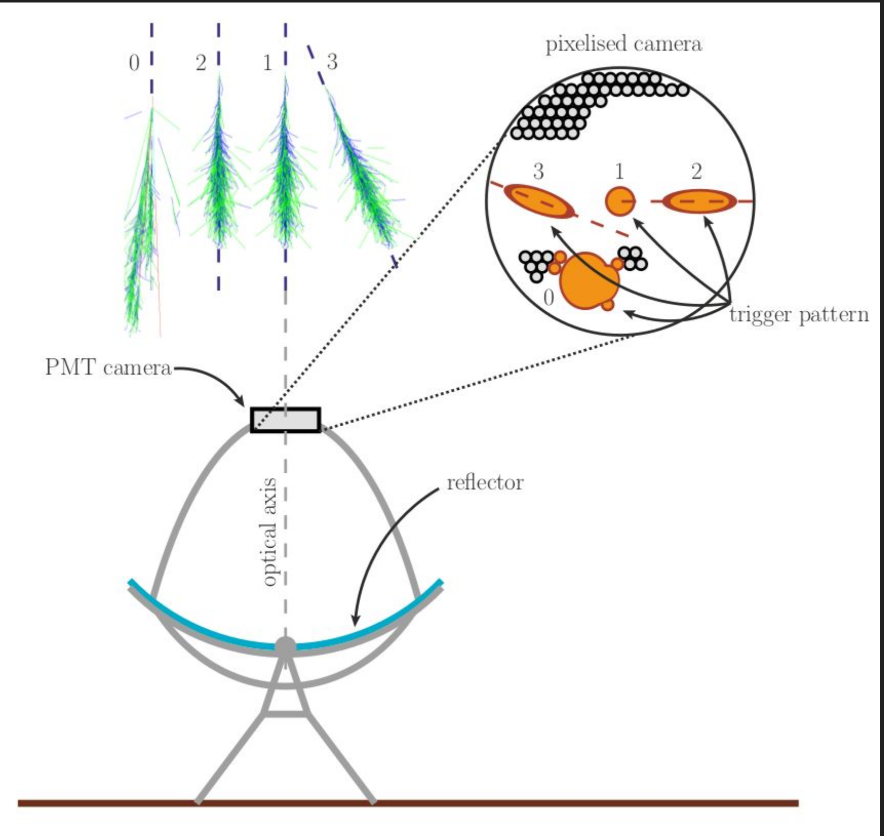


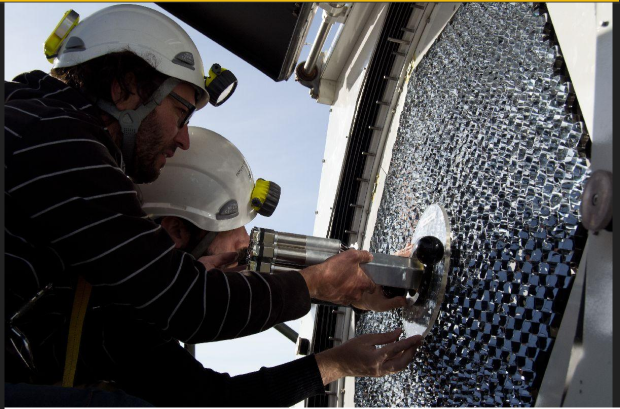
Cherenkov Technique: Air-showers

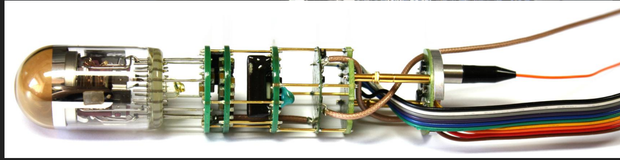




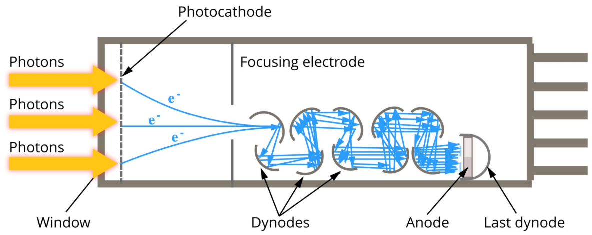
Cherenkov Technique: camera



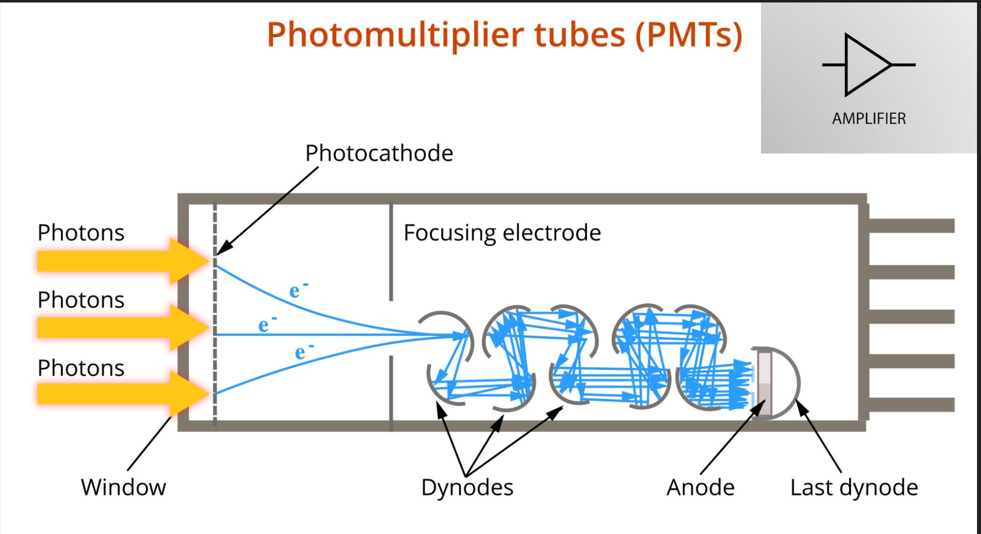




Photomultiplier tubes (PMTs)

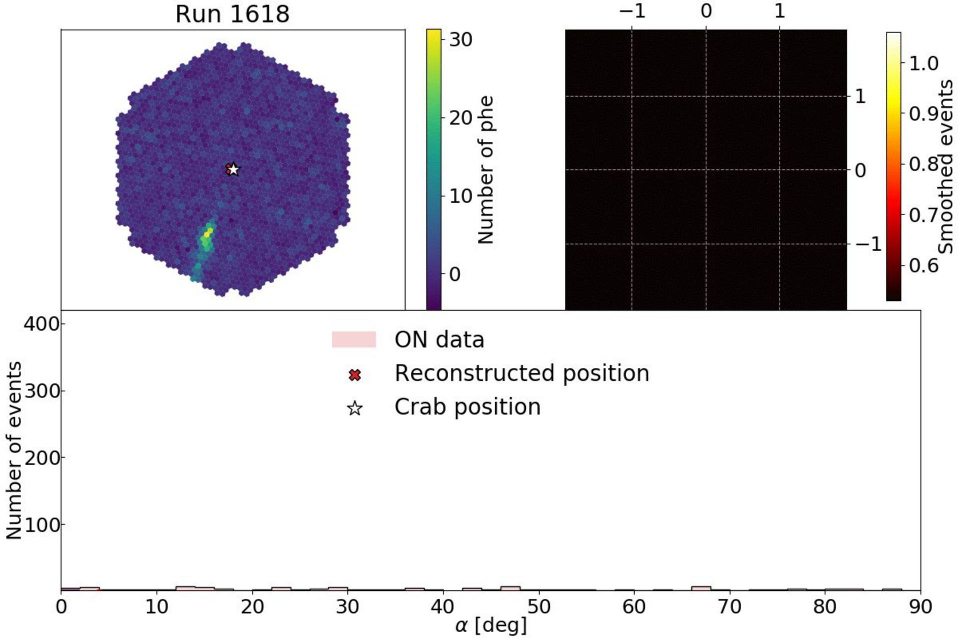


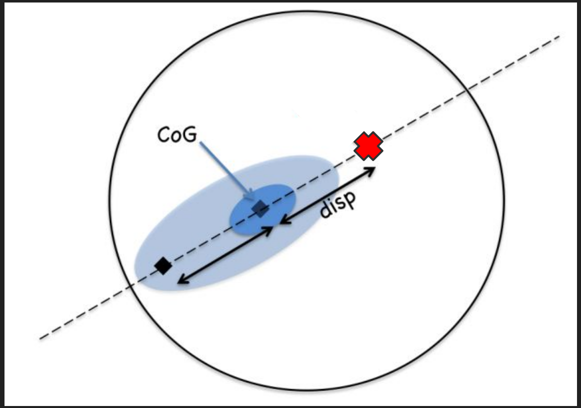
Cherenkov Technique: camera



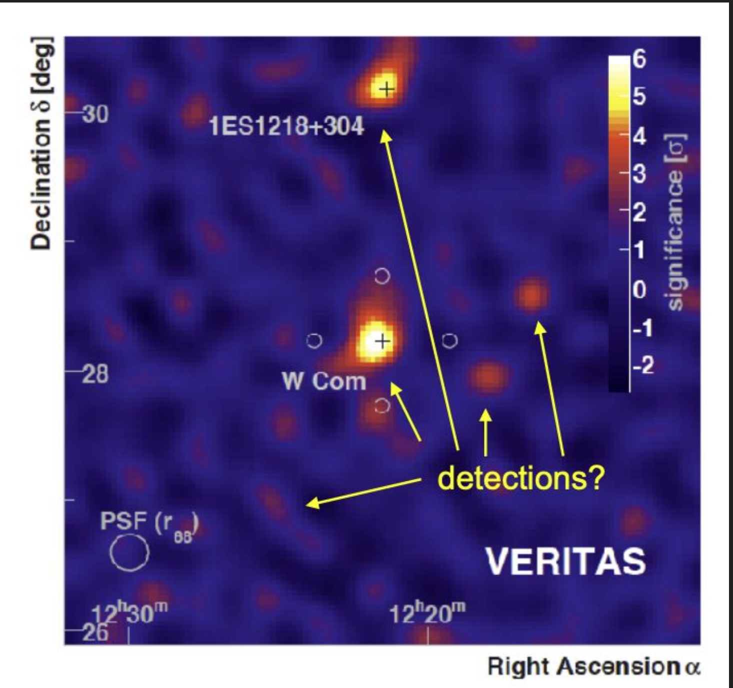
★ Question: Which similar device have we seen before?
  ○ Hint: this morning's lecture

Cherenkov Technique: Point-source Detection

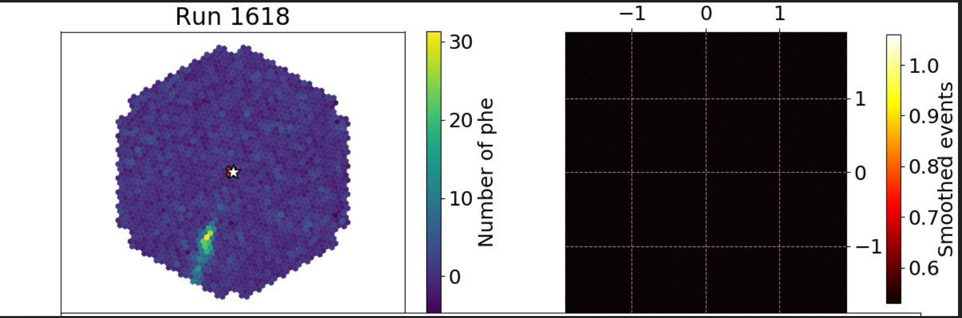


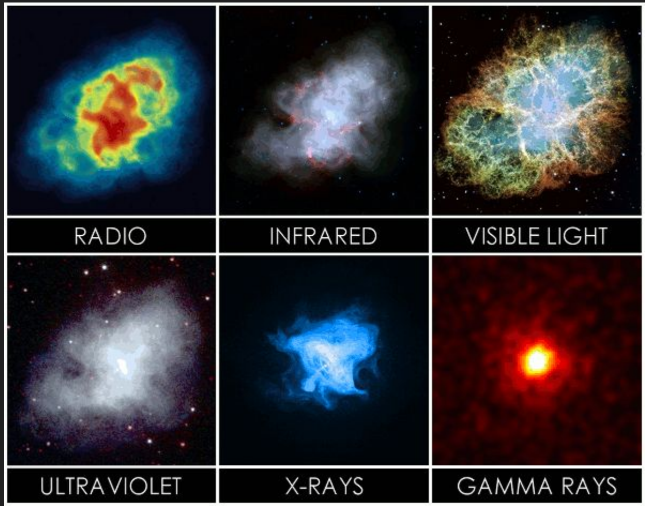


Cherenkov Technique: Point-source Detection



Cherenkov Technique: Point-source Detection





Angular resolution comparison

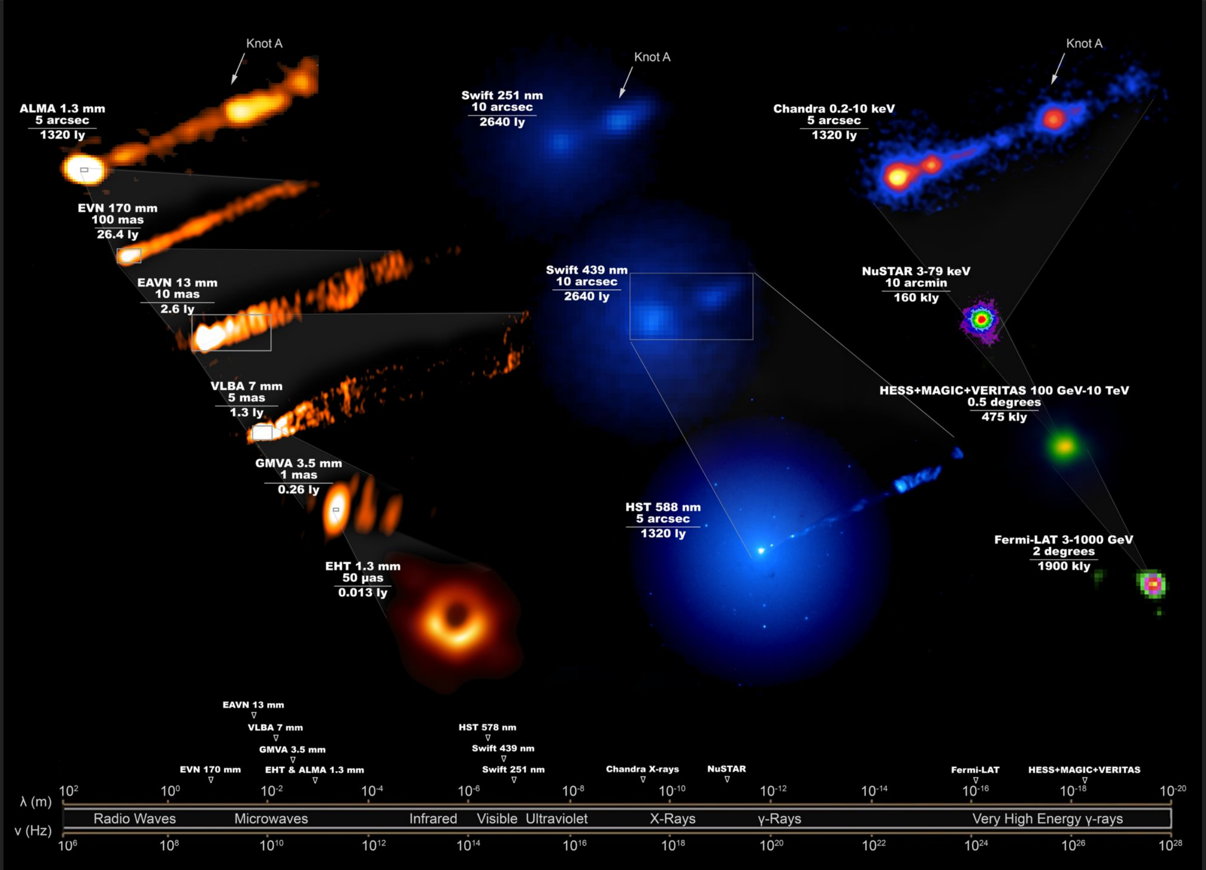
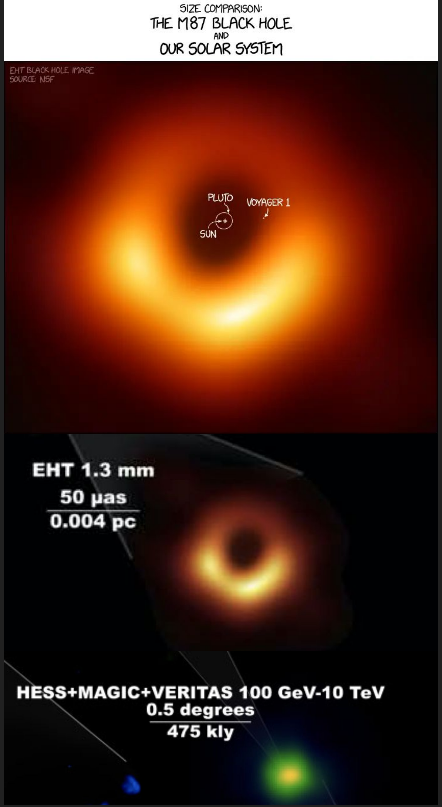


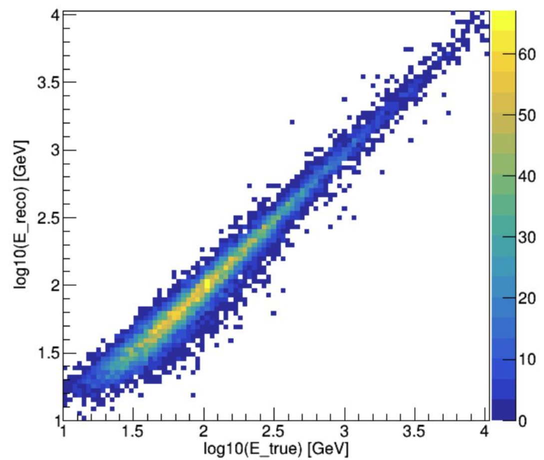
Image Credit: The EHT Multi-wavelength Science Working Group; the EHT Collaboration; ALMA (ESO/NAOJ/NRAO); the EVN; the EAVN Collaboration; VLBA (NRAO); the GMVA; the Hubble Space Telescope; the Neil Gehrels Swift Observatory; the Chandra X-ray Observatory; the Nuclear Spectroscopic Telescope Array; the Fermi-LAT Collaboration; the H.E.S.S collaboration; the MAGIC collaboration; the VERITAS collaboration; NASA and ESA. Composition by J. C. Algaba

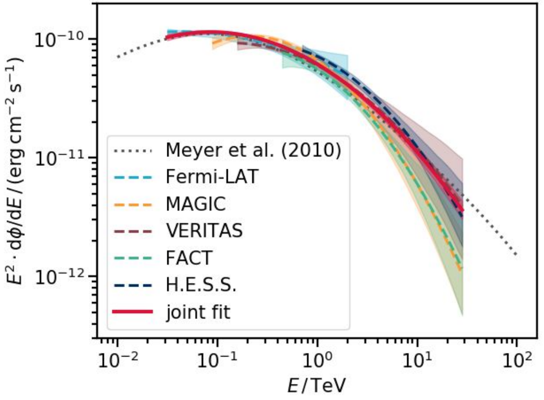


1 degree = 3,600,000,000 µas

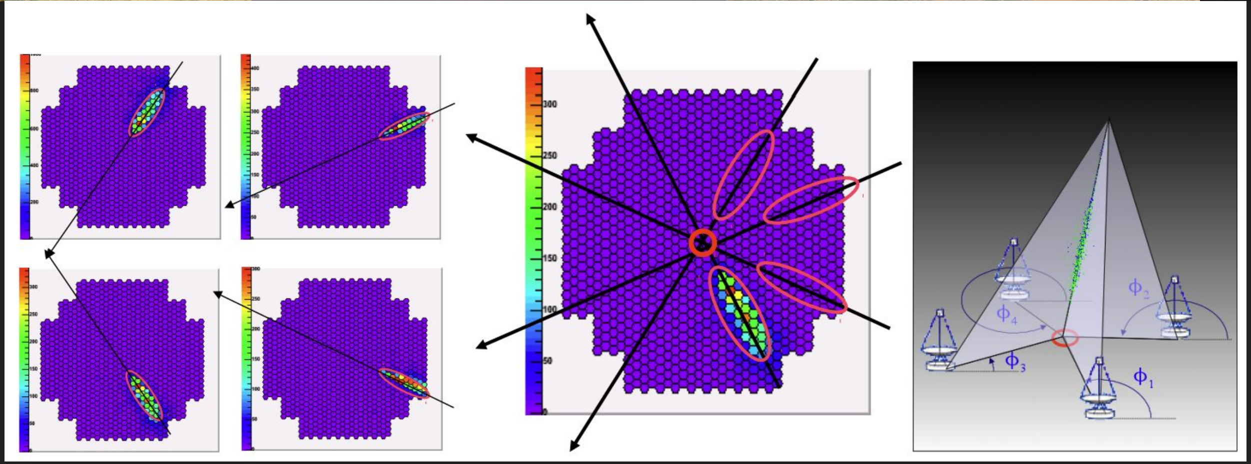
Cherenkov Technique: Energy Reconstruction ("Spectroscopy")

- ★ Based on simulation
- ★ Several methods:
  - ○ Lookup table (Energy ~ number of photons)
  - ○ Template (Energy ~ shower image)
  - ○ Other fancy ML techniques (Random Forest, Boosted Decision Tree, CNN)
- ★ Obtain flux and spectral energy distribution as a function of energy
- ★ High systematic uncertainty (~10%)

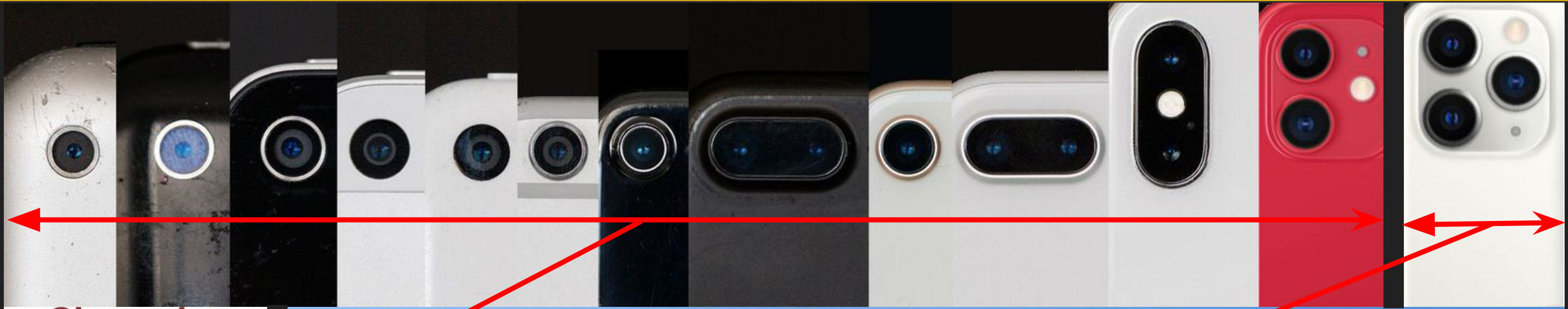


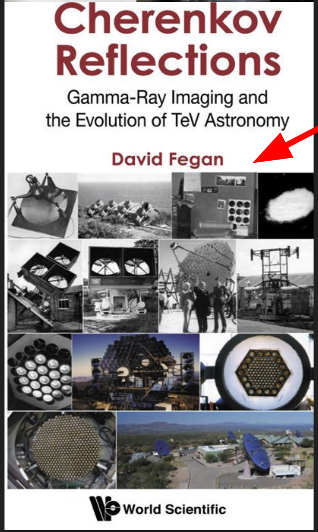


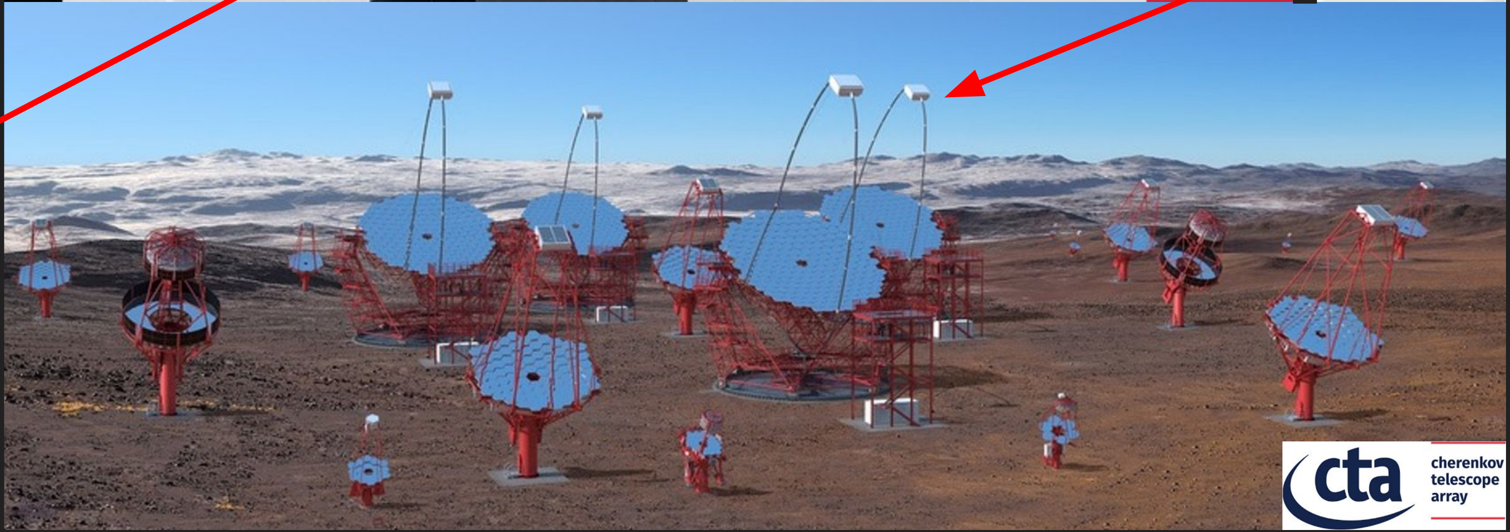
Cherenkov Technique: current-generation telescopes



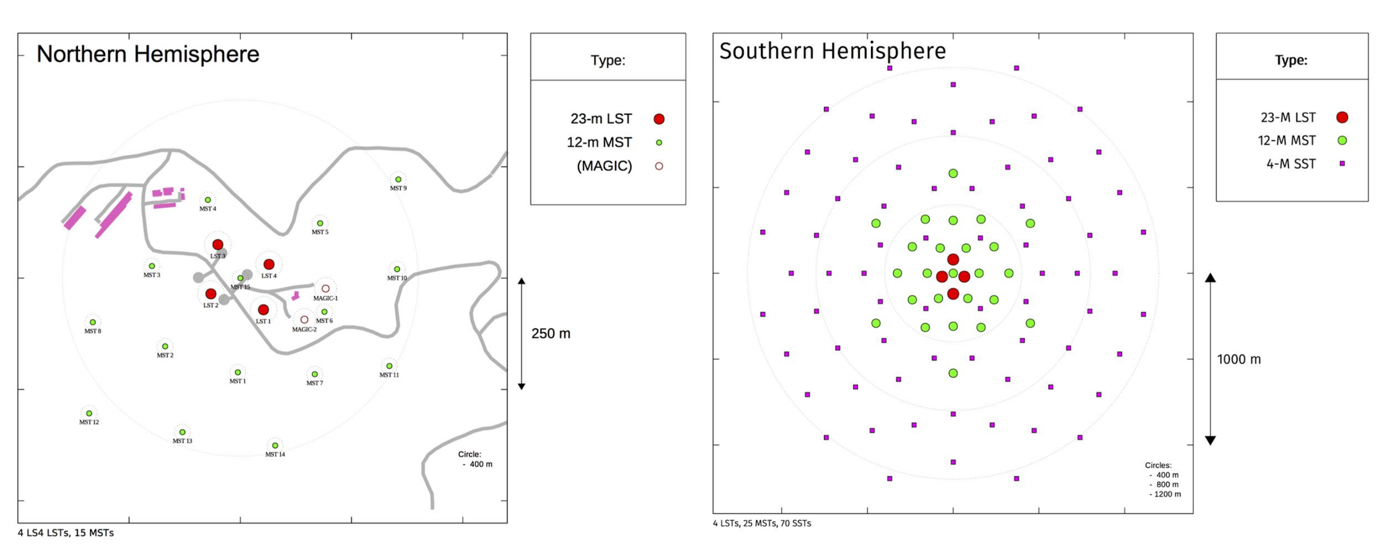
More cameras, more pixels!

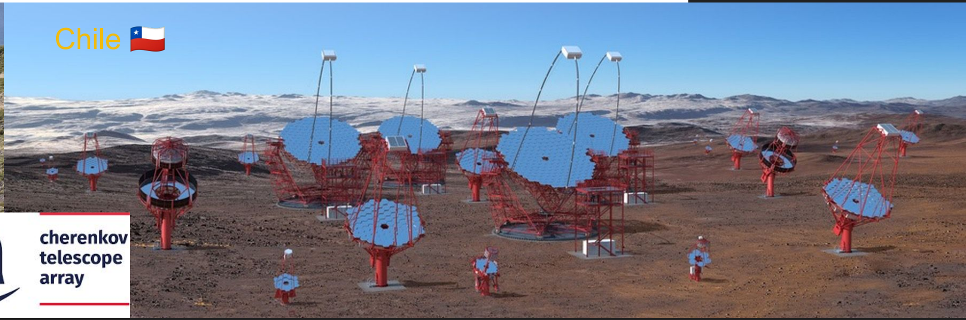






More telescopes!

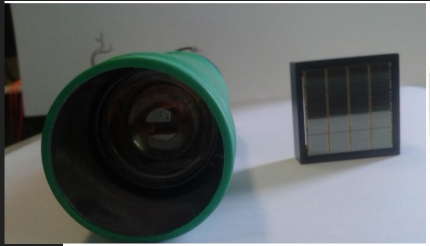




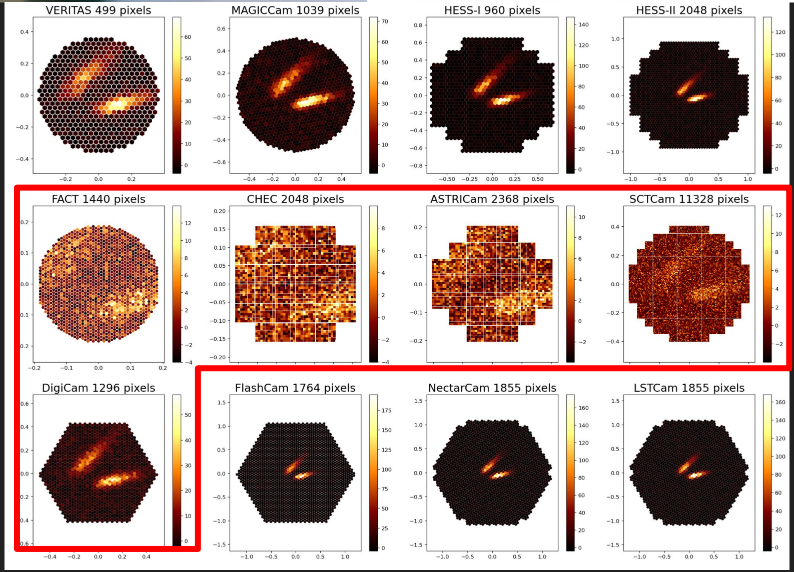
cta cherenkov telescope array

From tubes to silicon sensors

PMT vs SiPM



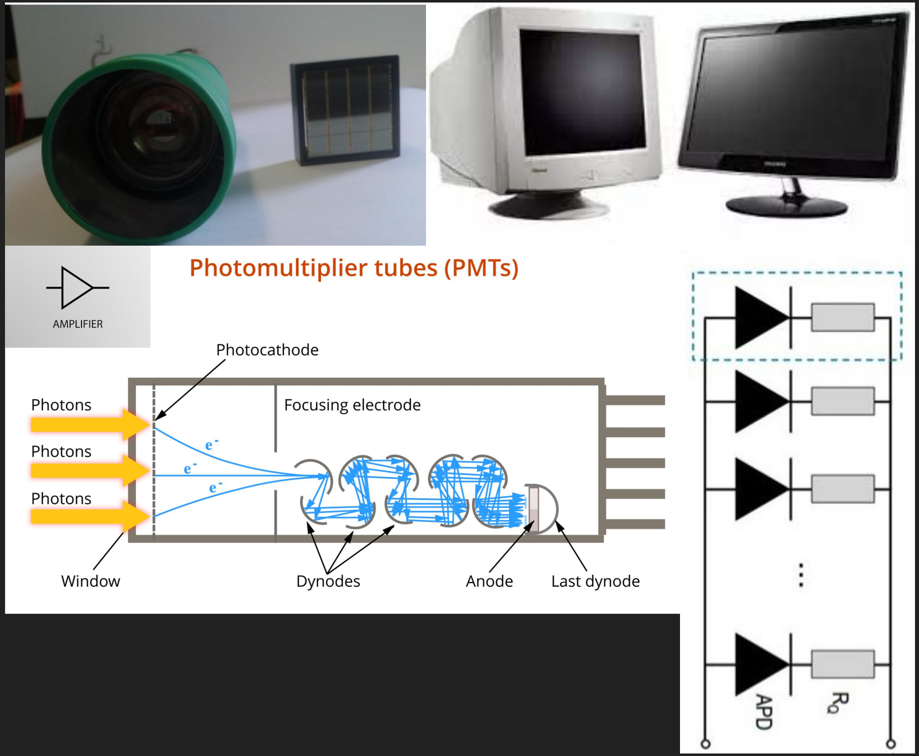
CRT vs LCD



- ★ Some large cameras still use PMTs:
  - Improved detection efficiency
  - Higher pixel density

From tubes to silicon sensors

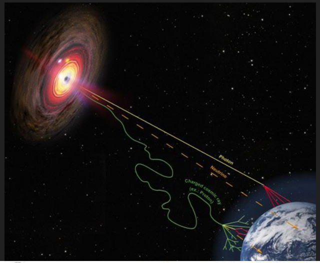
PMT vs SiPM CRT vs LCD

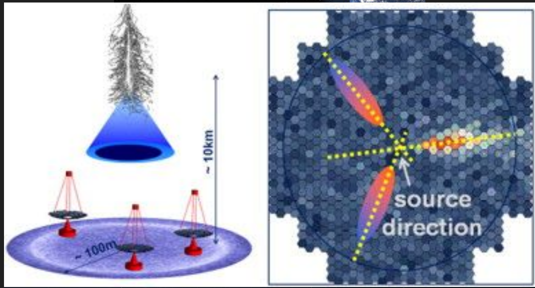


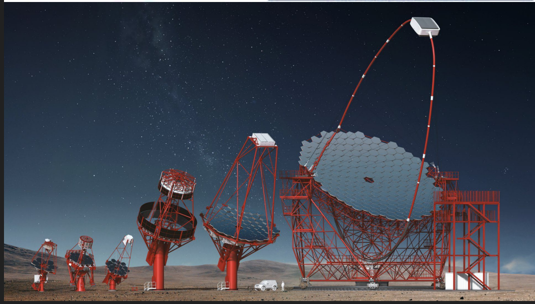
- ★ Some large cameras still use PMTs:
  - Improved detection efficiency
  - Higher pixel density
- ★ Silicon PhotoMultiplier:
  - Silicon is relatively “cheap”
  - More compact and robust
  - Lower operating voltage
  - Higher pixel density
  - Comparable detection efficiency
  - Single-photon resolution

Cherenkov Telescopes are funky...

- ★ Look for energetic gamma-ray sources, indirectly:
  - Captures “images” of air-showers
  - Reconstructs source’s <u>location</u> based on <u>orientation</u> of images
  - Reconstructs <u>energy spectra</u> based on <u>shape</u> of images
- ★ Do not quite resemble conventional optical telescopes:
  - Final images are low-res, effectively point-source
  - Fast camera (PMT, ns-timescale) instead of CCD (second-timescale)
  - Large segmented mirrors, no dome, come in array



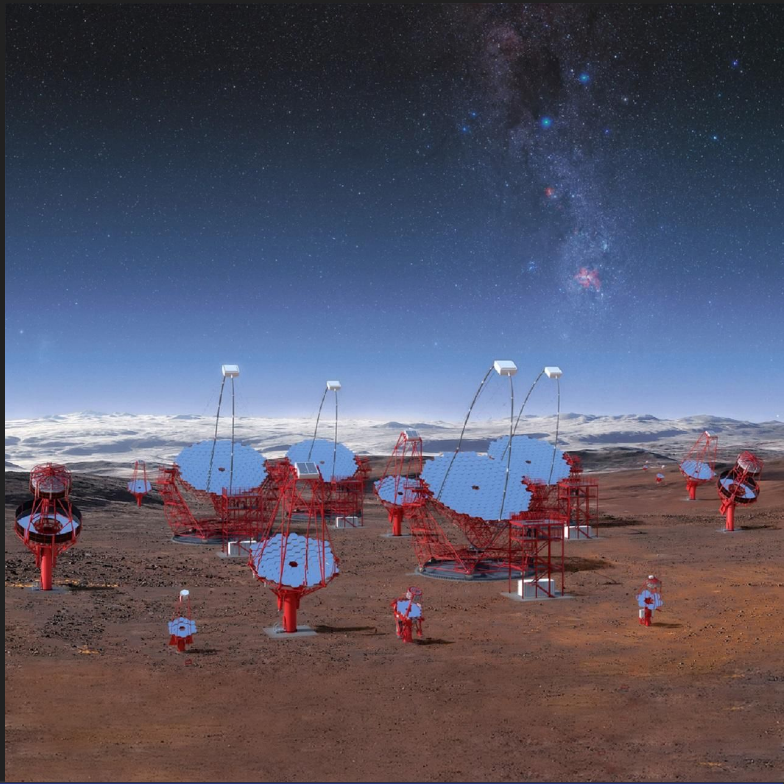




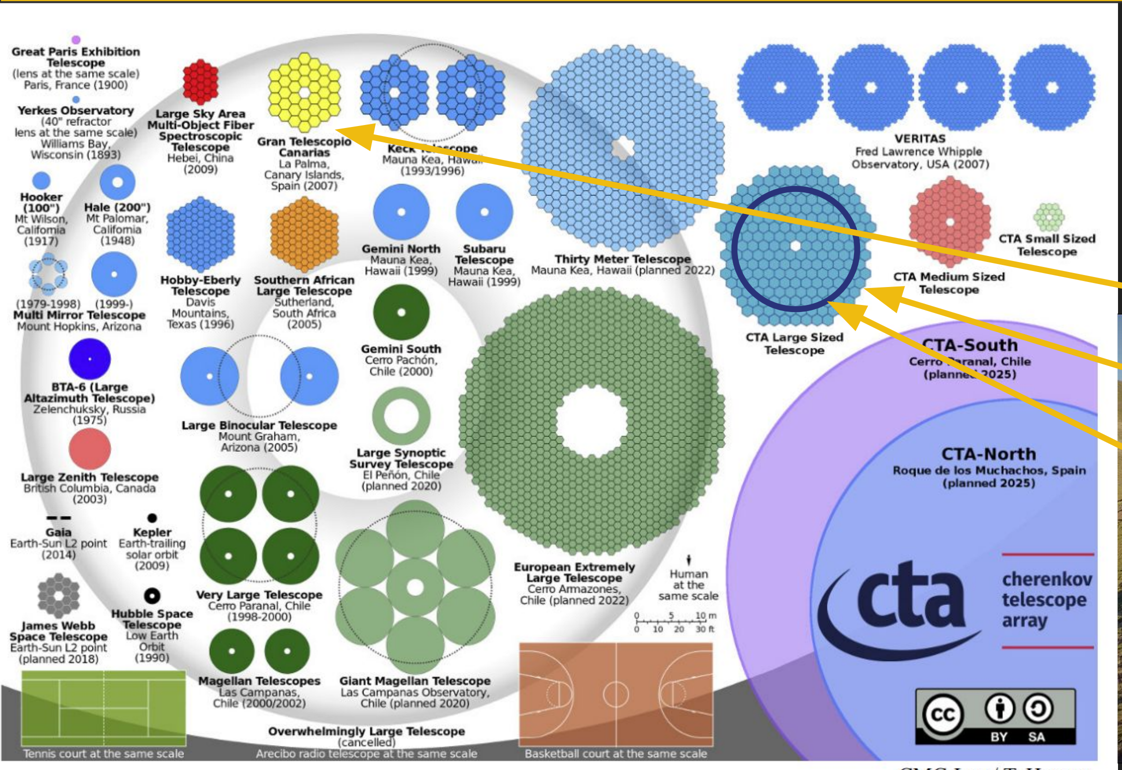
Pause for Q&A

Questions on Cherenkov Technique?

Question: why are the mirrors made of many individual panels?

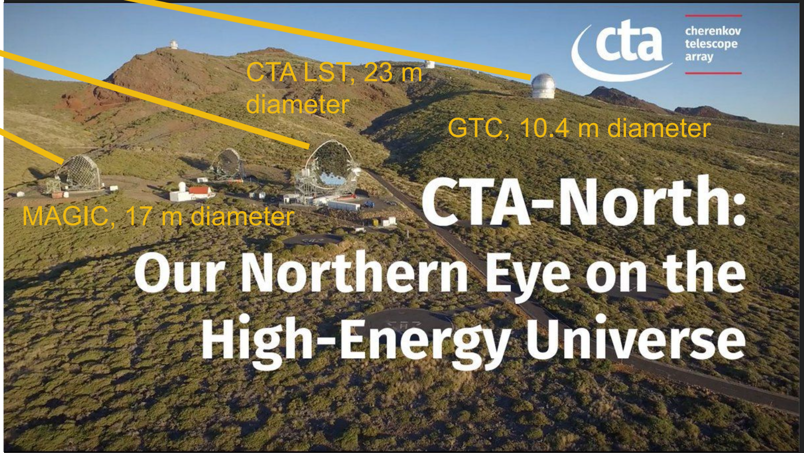


Cherenkov Telescopes are still optical telescopes



CMG Lee / T. Hassan

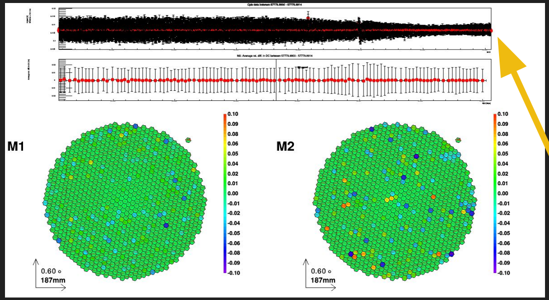
- ★ Modest mirror quality but large collecting area.
- ★ Effective “photon bucket”.
- ★ Fast (ns) PMT as detector, vs. slow (ms) CCD in conventional optical telescopes.



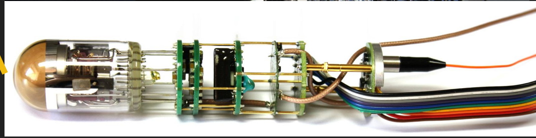
Mirror size of optical telescopes

One-pixel telescope

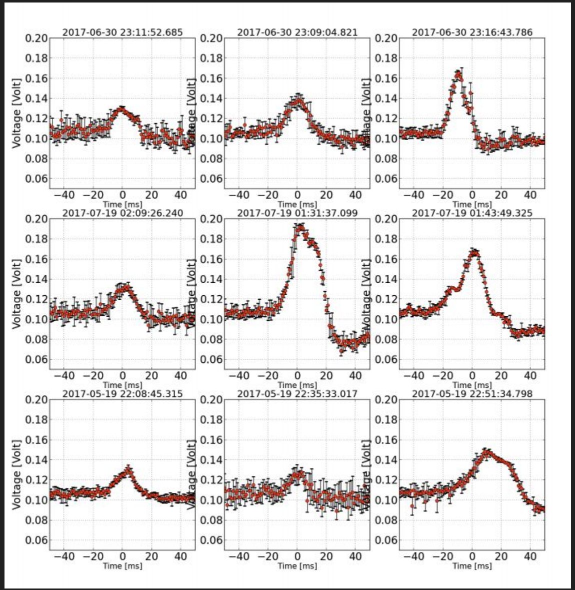
- ★ First Cherenkov “telescope” in 1952:
  - A dustbin, a 25-cm parabolic mirror, a PMT, some simple electronics
- ★ Modified central sensors to read out <u>simultaneously</u> optical signal at 10 kHz rate (0.1° FoV):
  - Sensitive to sub-ms optical pulses (<u>continuous sampling</u> for optical observation)



MAGIC CPix

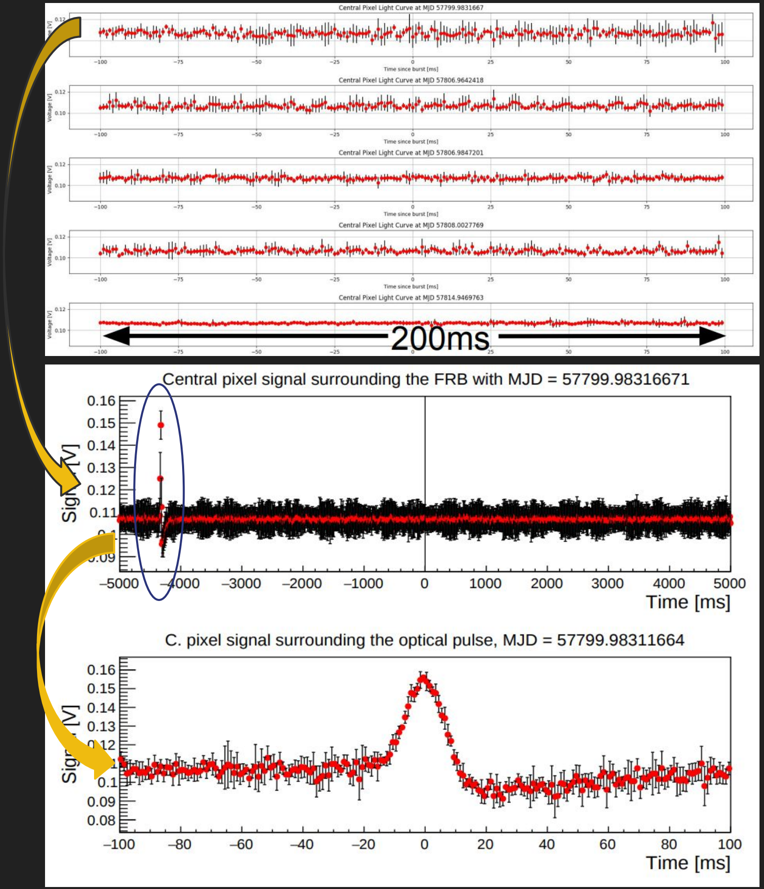


One-pixel telescope: meteors

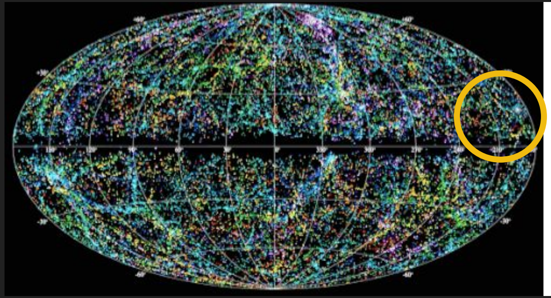


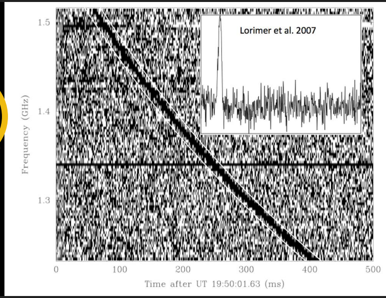
Some examples of bright isolated optical pulses due to meteors (MAGIC)

One-pixel telescope: Fast Radio Bursts

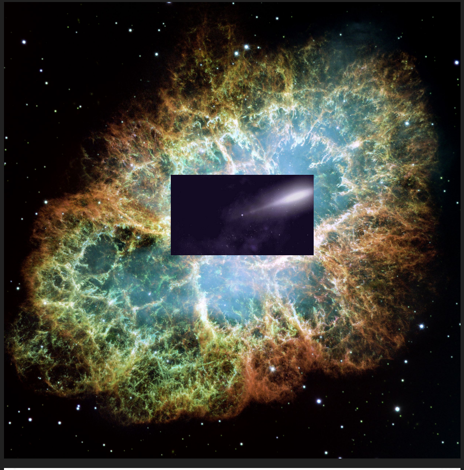


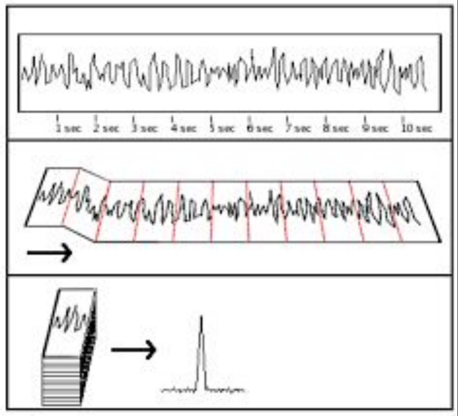
- ★ FRBs: transient radio events, bright and short
- ★ Unknown origin, tentative candidate: magnetar
- ★ FRB 121102: first repeating FRB
- ★ MAGIC observed in both optical and gamma-ray
- ★ An optical burst was detected ~4 seconds before the first radio burst, most likely a background event

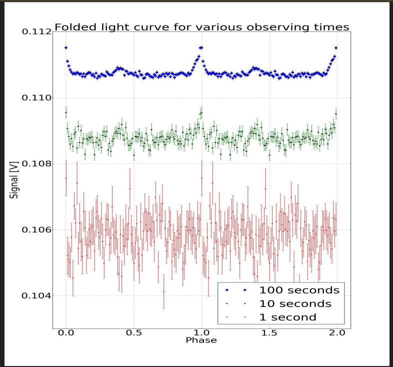


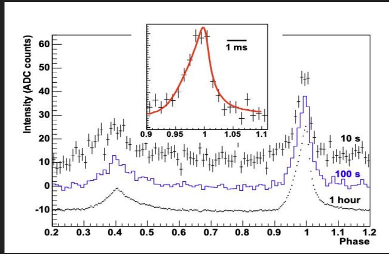


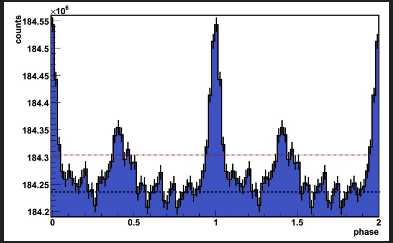
One-pixel telescope: optical pulsar











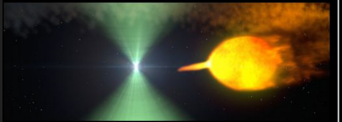
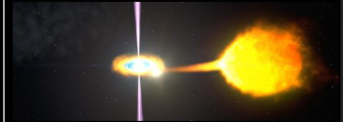
CPix on 17m MAGIC

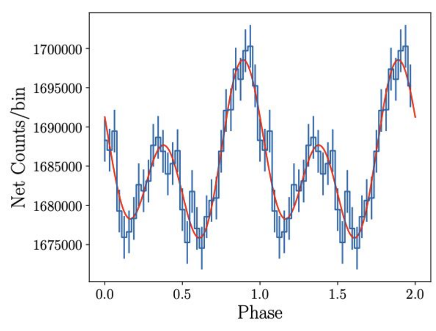
Optical pulsar camera (no longer used)
on 12m H.E.S.S.-I

Current monitor on 12m VERITAS

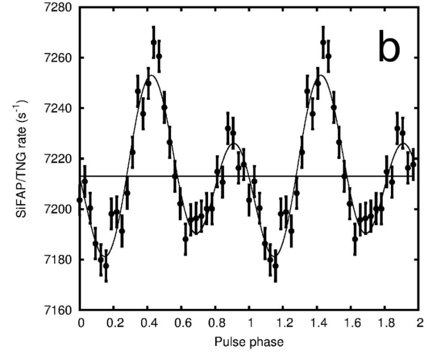
One-pixel telescope: transitional millisecond pulsar

Optical pulsation from PSR J1023+0038 has been detected by several instruments

Radio Pulsar State	Intermediate Disc State
	
• Observed radio/gamma-ray pulsar. • Likely radio eclipses. • Lots of orbital timing noise. • Modulation of X-rays at orbital period (shock).	• No visible radio pulsar (off?). • Increased optical, X-ray, and gamma-ray brightness. • Double peaked optical emission lines. • Flat-spectrum radio continuum source (jet?). • No X-ray orbital modulation. • X-ray dropouts and flares.



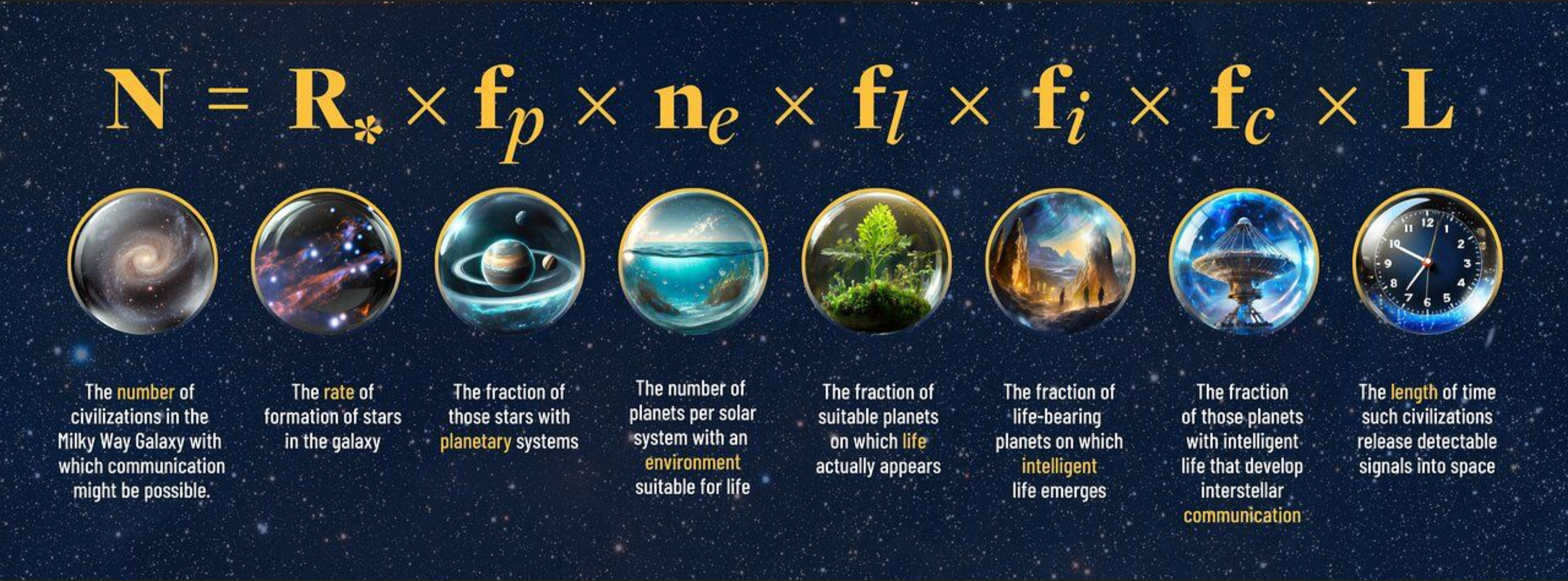
Aqueye+ on 1.8m Copernicus telescope Asiago



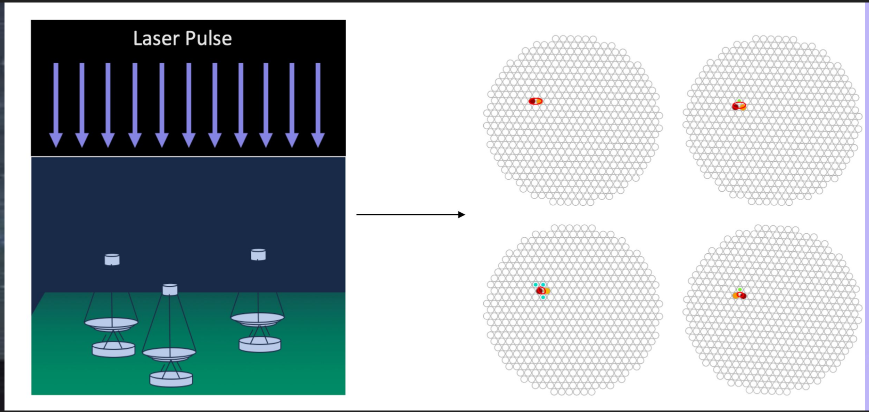
SiFAP2 on 3.58m TNG

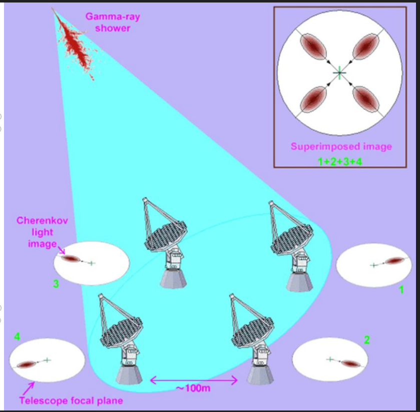
Many-pixel telescope: Optical SETI

- ★ Question: does alien exist?
  - ○ Raise your hand if you think alien exists



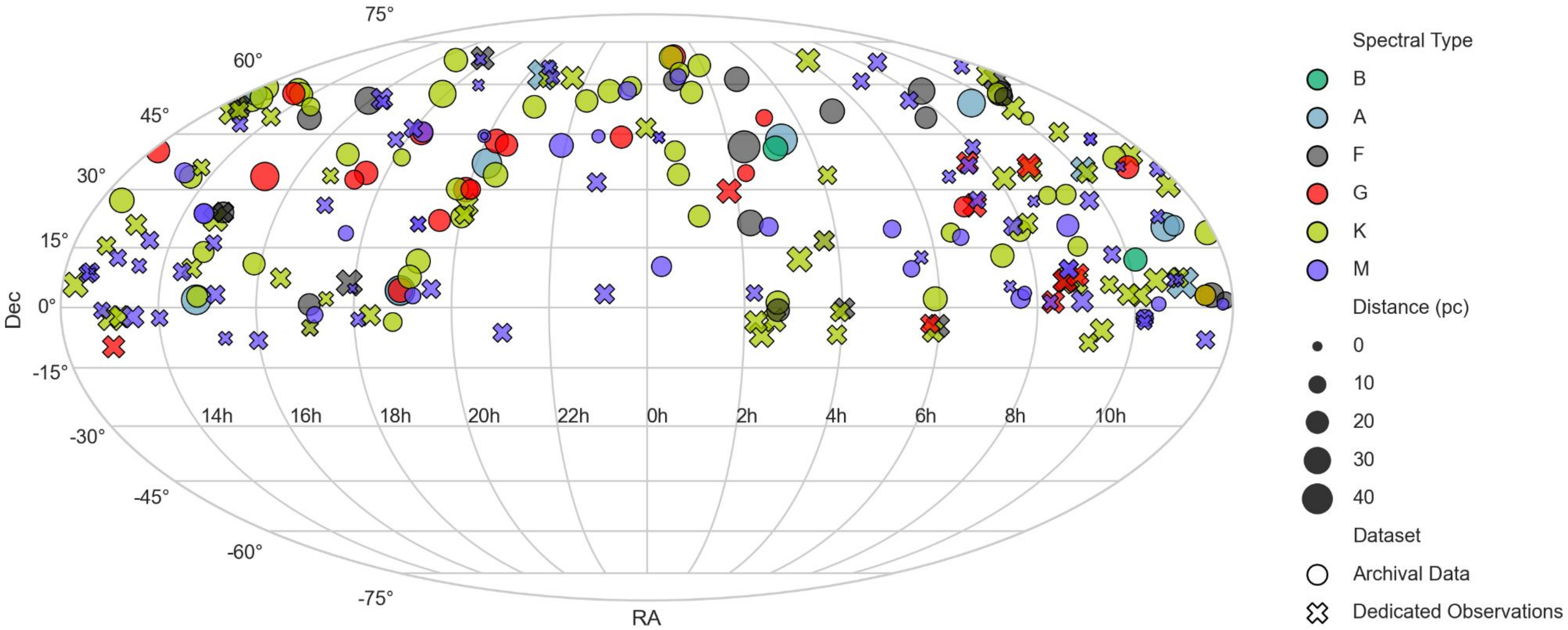
Many-pixel telescope: Optical SETI





- ★ Laser pulses should appear point-like with equal intensity and in the same position in each camera.
- ★ Previous search for laser from KIC 8462852 (see arXiv: 1602.00987).

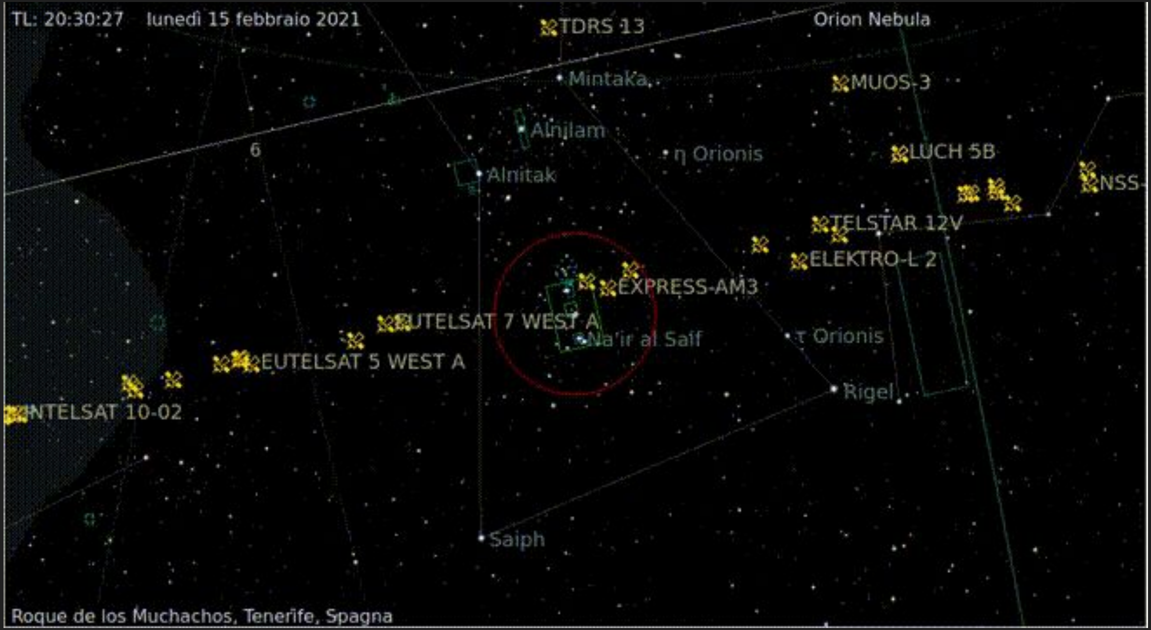
Many-pixel telescope: Optical SETI

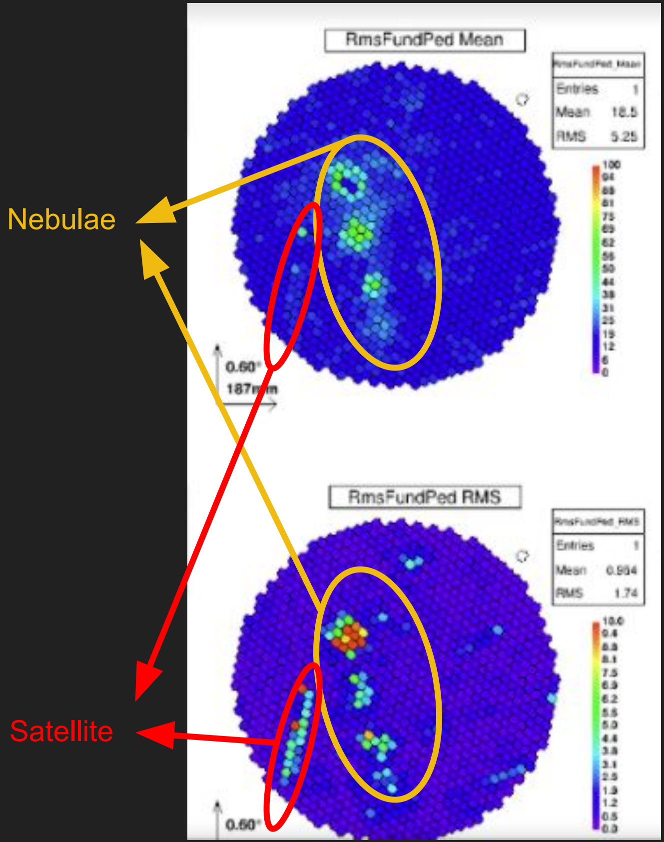


2019 - 2020 observational and archival campaign

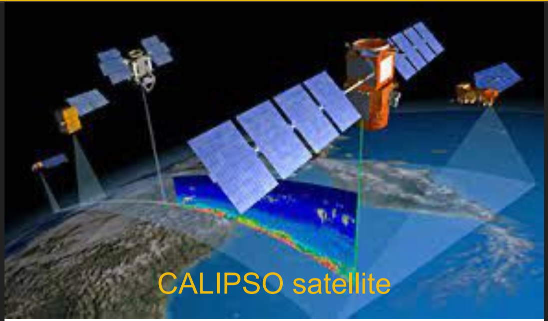
Many-pixel telescope: technosignatures search

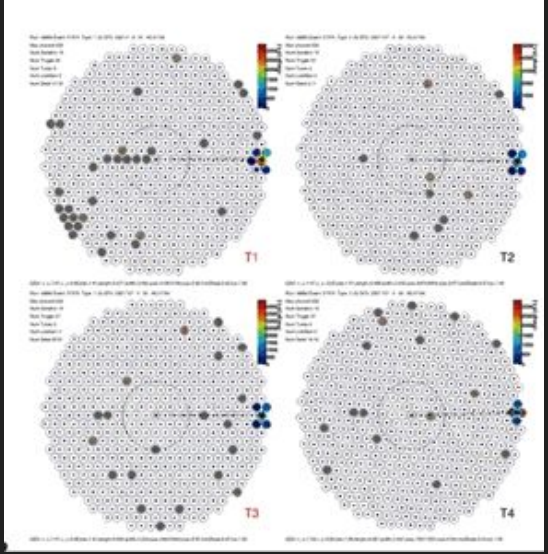
- ★ It's a jungle up there, really.
- ★ Sensitive enough to detect lidar laser from some satellites.
- ★ Some fantastic applications

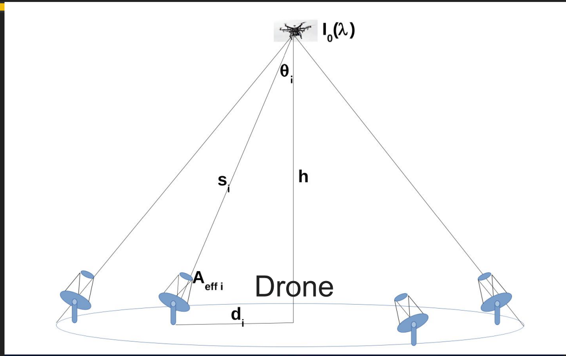


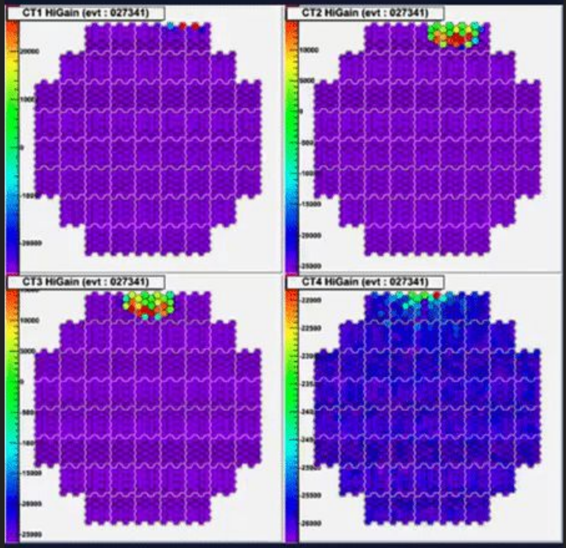


Many-pixel telescope: technosignatures search



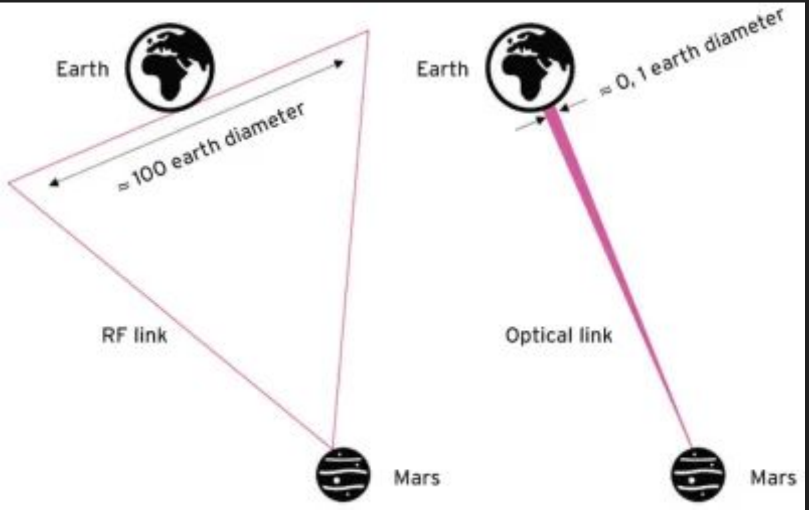


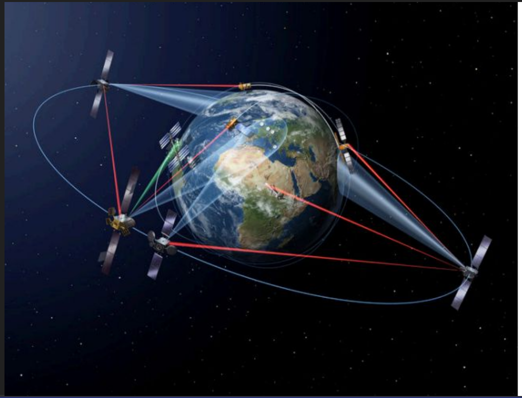


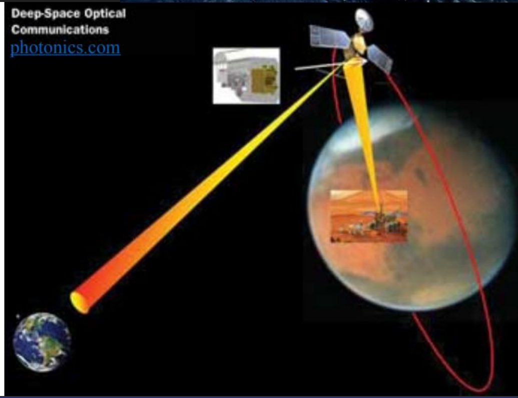


Many-pixel telescope: technosignatures detection

- ★ Optical tracking of orbiting objects
- ★ Optical Wireless Communication
- ★ Secure Quantum Communication
- ★ Question: In what way(s) is optical communication is better than radio communication?

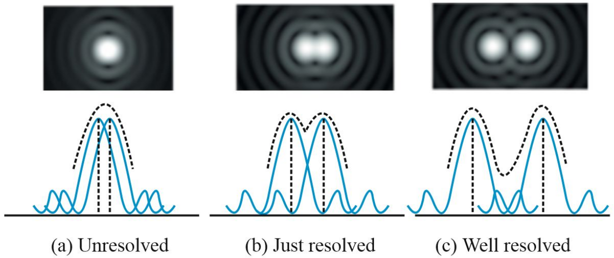






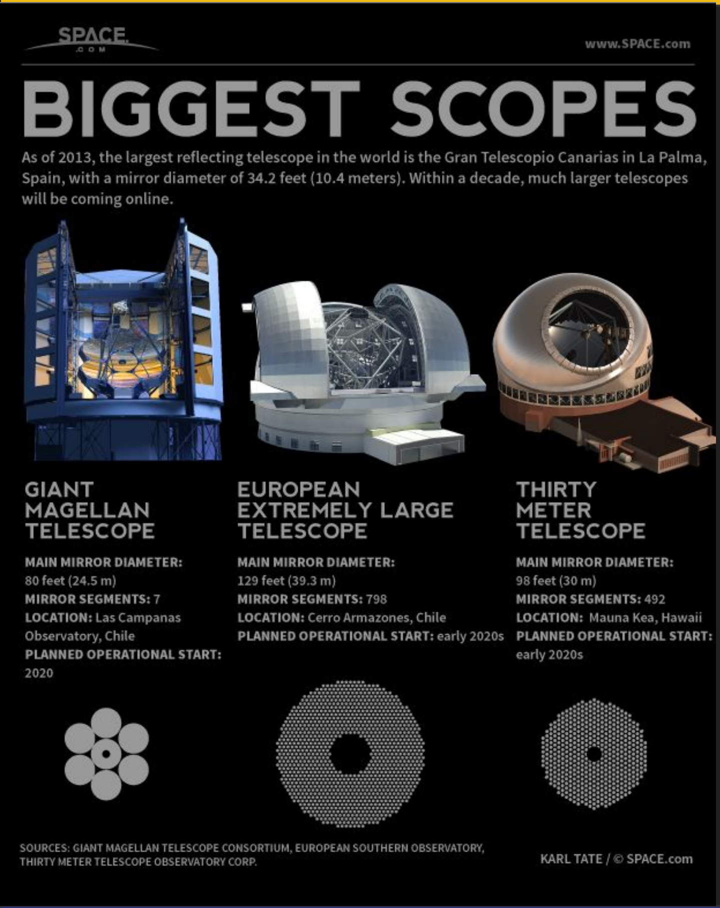
Many-telescope telescopes: angular resolution limit

$$\theta = 1.22\frac{\lambda}{D}$$

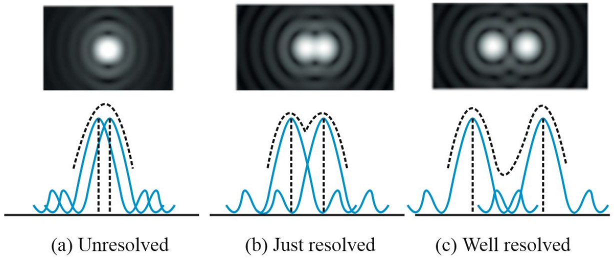


Rayleigh's criterion for resolution of objects.

Many-telescope telescopes: angular resolution limit



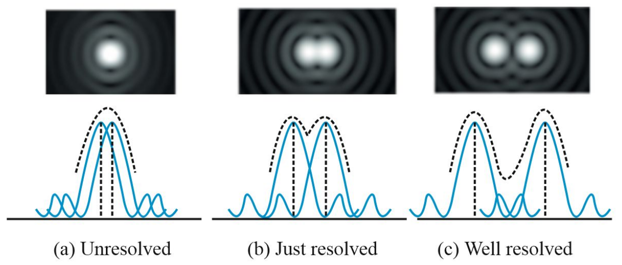
$$\theta = 1.22\frac{\lambda}{D}$$



Rayleigh's criterion for resolution of objects.

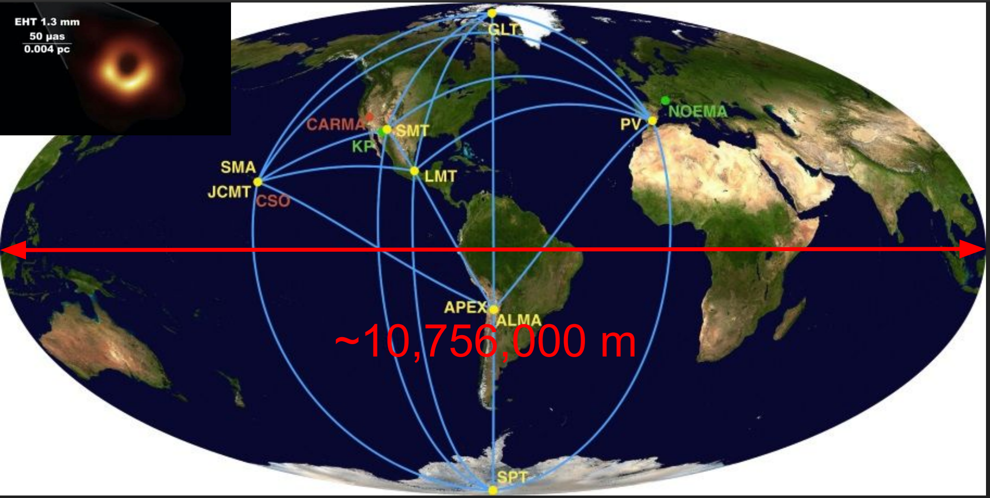
Many-telescope telescopes: optical interferometry

$$\theta = 1.22\frac{\lambda_{\text{mm}}}{D}$$

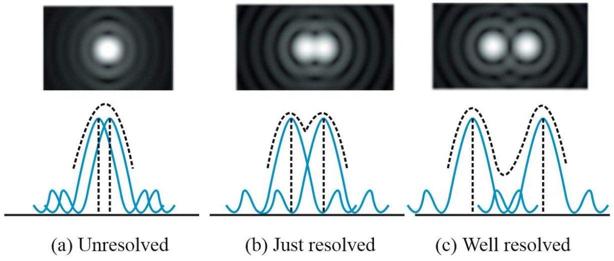


Rayleigh's criterion for resolution of objects.

Many-telescope telescopes: optical interferometry

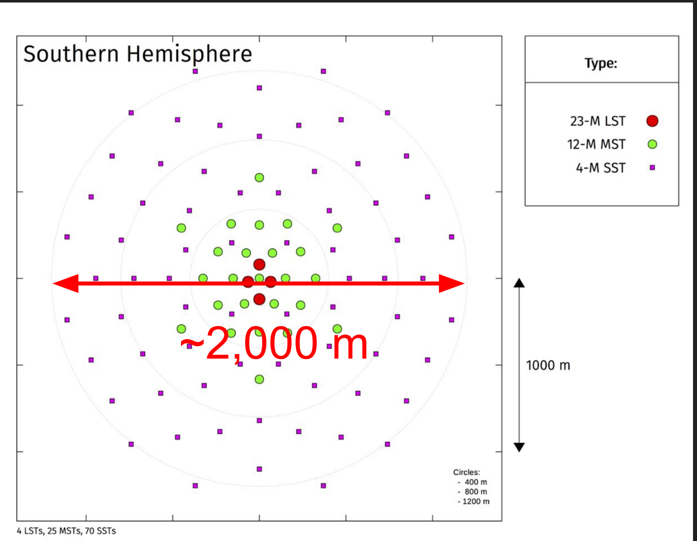


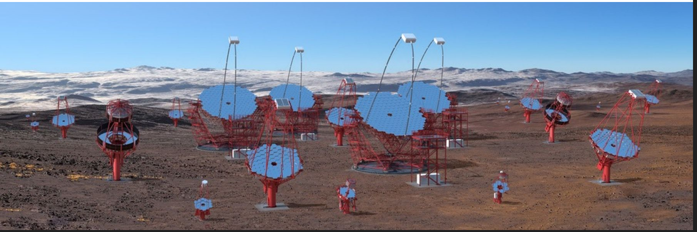
$$\theta = 1.22\frac{\lambda_{mm}}{D}$$



Rayleigh's criterion for resolution of objects.

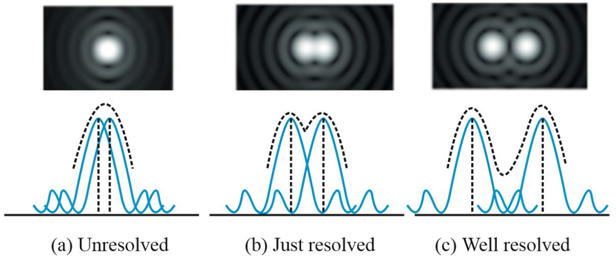
Many-telescope telescopes: optical interferometry





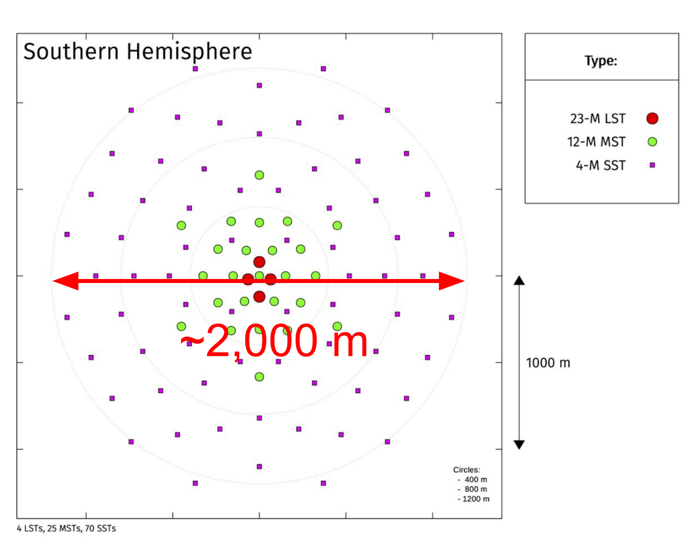
1mm ~ 2200 x 450 nm

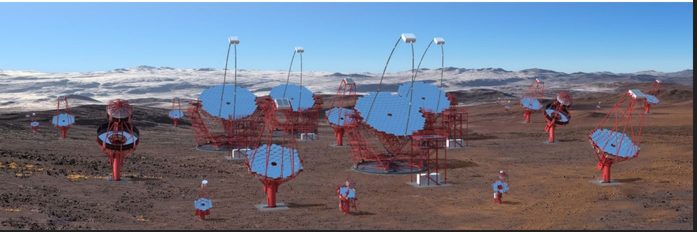
$$\theta = 1.22\frac{\lambda_{nm}}{D}$$

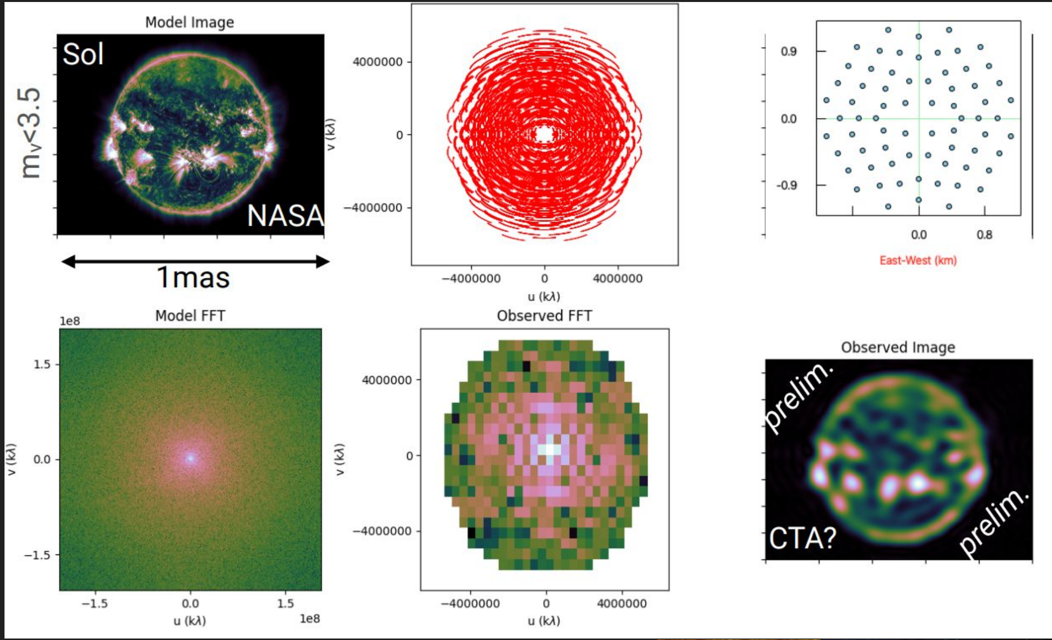


Rayleigh's criterion for resolution of objects.

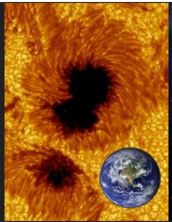
Many-telescope telescopes: optical interferometry

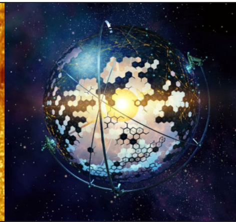






- ★ Simulated 2h observation with CTA-South SSTs for a <1mas “solar-like” star
- ★ sub-mas resolution => New windows into SETI (exoplanet/Dyson sphere imaging)





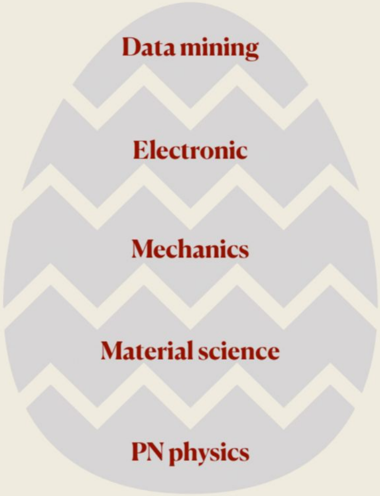
Take-home messages

- ★ Cherenkov telescopes are unconventional “particle detector”:
  - Large optical dishes for gamma-ray astronomy
  - Capture airshower images to “measure” properties gamma-ray sources
  - Diverse range of targets
  - Applicable to OSETI & flying objects searches
- ★ Complementing “traditional” optical astronomy through multiple aspects:
  - Time-domain: from ns to ms and second
  - Multiwavelength: gamma and optical
  - Pixel => Telescope => Array

Q&A

Thank you for your attention!

Particle detector is a *instrument (kind of classical device)* to detect, to track, and to identify the *particles (superposition of quantum states)*



A complicate, interdisciplinary field

"The real voyage of discovery consists, not in seeking new landscapes, but in having new eyes"

~Marcel Proust~

Good luck!

From Son Cao

And can be a risky business (sometimes…)

How to vaporize a solid piece of steel in a few seconds

Q&A

Questions on Cherenkov Technique?

Question: What will happen if you point the Cherenkov telescope to the sun?

- Hint: The power per m^2 of the Sun at the Earth's surface is approximately 1370 W/m^2

Cherenkov ring

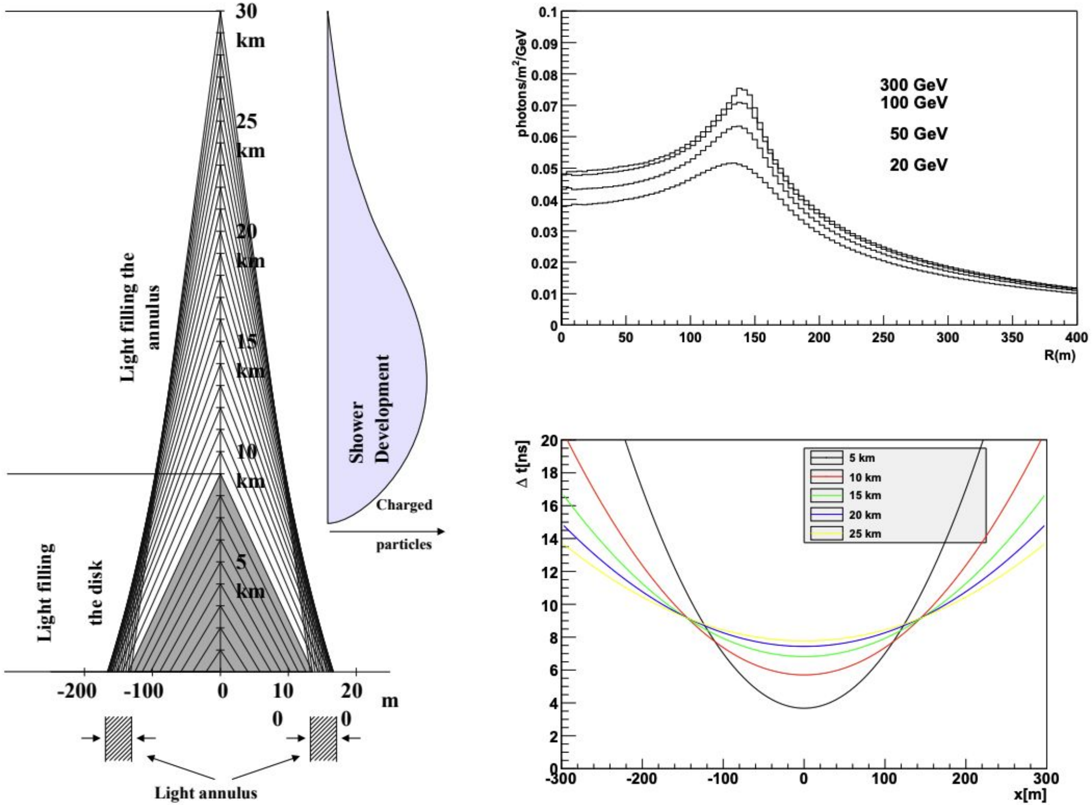
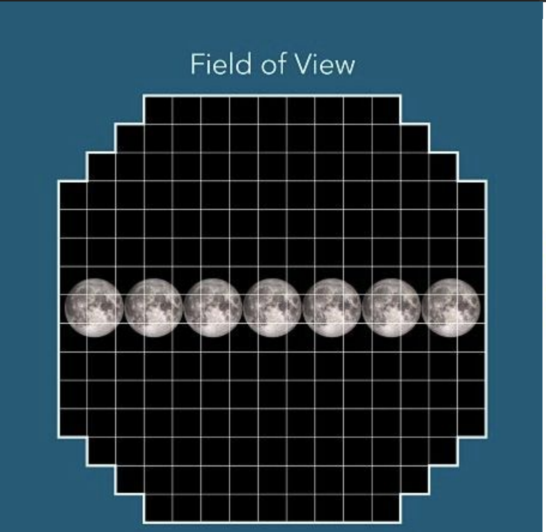
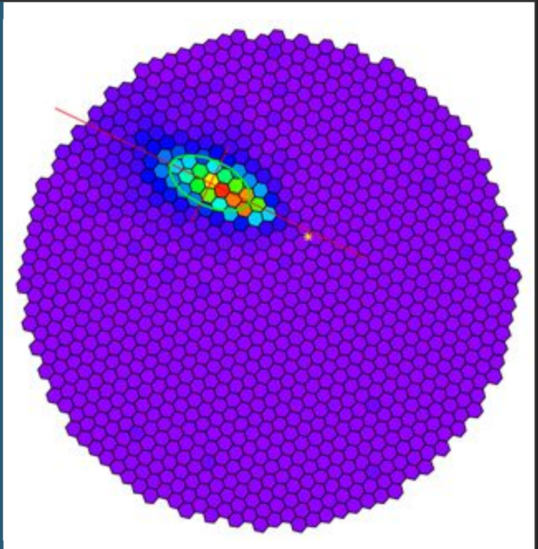


Figure 4. **Left:** Shower development. **Top Right:** Lateral profile of showers of different energies at sea level. **Bottom Right:** Time delay as function of lateral distance for various altitudes of emission.

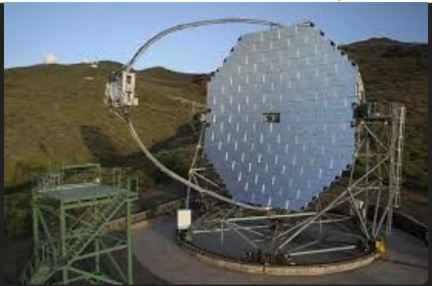
FoV comparison



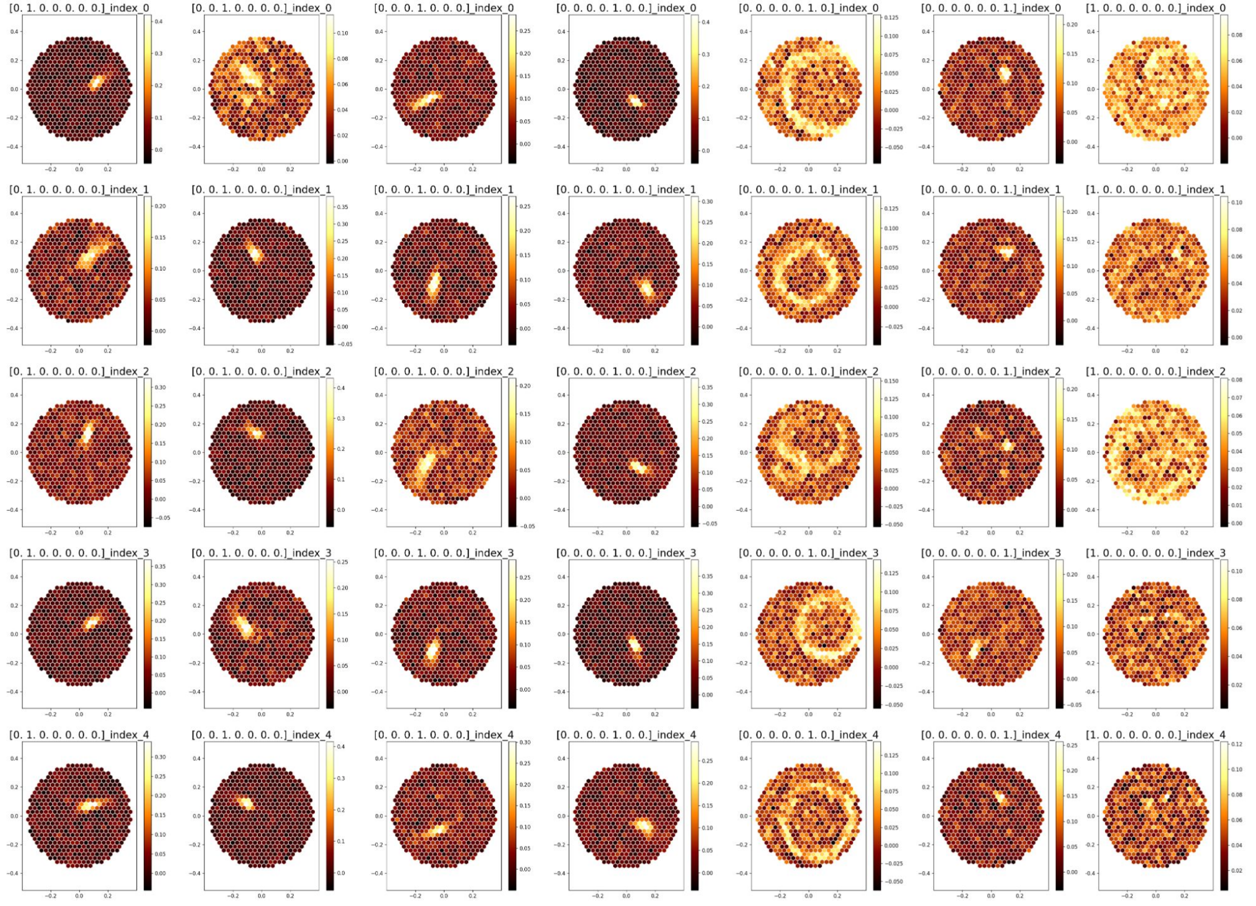
LSST: 3.5º FoV, 3.2 Gpixels



MAGIC, 3.5º FoV, 1 kpixels



Different showers type



SiPM

$$V_{bias} = V_{breakdown} + V_{Overvoltage}$$